3. RESULTS

One hundred and twelve patients (108 men and 04 women) were included in the study. The demographic characteristics of the participants are shown in Table **1** including the average age of 69.49 years ± 8.15, smoking of 53.21 ± 48 P / A on average, and an FEV1 of 42.55 ± 18.90 l / s. COPD was recorded at 22.5% as moderate, 43.6% as severe, and 33.9% as very severe. The average six-minute walk test was 273,79 ± 112, 79 m. BODE index was 5, 54± 2, 63. Of the 112 cases included, 21 had COPD associated with bronchiectasis (18.75%), as shown in Fig. (**1**).

Bronchiectasis occurred in the lower lobes in 15 cases (71.42%) and the upper lobes in 06 cases (28.58%). It was the cylindrical type of bronchiectasis in the majority of patients, *i.e.*, 17 (80.95%), saccular in 03 cases (14.28%), and mixed in only one case (04.77%). Lesions were bilateral in 16 patients (76.19%) and unilateral in 05 (23.81%) cases, as shown in Table **2**.

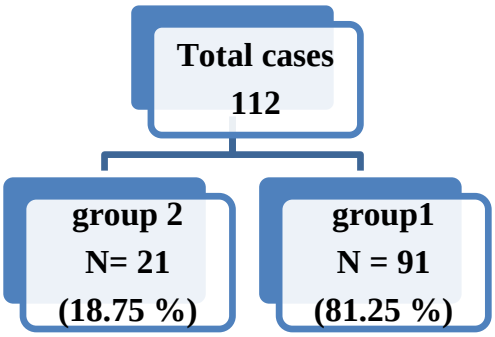


Fig. (1). Frequency of bronchiectasis in patients with COPD.

Table 1. Characteristics of the study population.

Setting	Features
Population	112
Age	69.49 Ans ± 8.15 ans
Sex	108 M - 4 F
Smoking	53, 21± 48 P/Y
BMI	20.49 ± 87 m^2/kg
COPD	Moderate 22.5%
	Severe 43.6%
	Very Severe 33.9%
FEV_1	42,55 ± 18,90%
BODE	5,54 ± 2,63
6MWT	273,79±112,79 m
I. Charlson	3,99 ± 2,32
mMRC	2.78 ± 0.79

Table 2. Characteristics of bronchiectasis on CT in group 2 (COPD associated with bronchiectasis).

Characteristics	N° (%)
$N_{O\ of\ Patients}$	21 (18.75)
Location	
Upper lobe	06 (28.58)
Lower lobe	15 (71.42)
Type	
Cylindrical	17(80.95)
Sacular	03(14.28)
Mixed	01(04.77)
Bilaterality	
Bilatéral	16(76.19)
Unilateral	05(23.81)

Table 3. Characteristics of bronchiectasis in both groups.

Characteristics	Goup 1	Goup 2	P-value
N_O	91	21	-
Age	70.69y	68.28y	0.378
Sex			
Male	87	21	-
Female	04	-	
Smoking	50, 11P/Y	56, 47P/Y	0.249
Charlson index	3.75±7.44	4.26±2.98	0.412
FEV_1	48.7±6 l/s	40.2±3l/s	0.017
mMRC	3±14	4±66	0.031
Purulent sputum	48%	64%	0.029
Anthonisen Index	-	-	0.521
Type 1	13 (10.83%)	44 (48.35%)	-
Type 2	39(42.8%)	04(3.33%)	-
Type 3	08(8.8%)	04(3.33%)	-
Length of hospital stay	08 days	12 days	0.014
SGQR total	34,51 ±14,23	46,11±74.39	0.032
CRP	25	38	0.001
Exacerbations in previous year	0.98	1.48	0.041

Table **3** shows the analysis of the characteristics of COPD in the two groups (G1: 91 cases of COPD without bronchiectasis and G2: 21 cases associated with bronchiectasis).

The mean age was 70.69 ± 3.37 y in G1 and 68.28 ± 9.14 y in G2. Regarding sex, only 04 female cases were found, all in group 1. The average smoking variable was 50,11 ± 80 P/Y in G1 and 56,47 ± 37P/Y in G2. Average tobacco consumption, Charlson's index, sputum purulence, FEV1, mMRC, mean hospital stay, SGQR, CRP, and exacerbations during the past year were, respectively: 3.75 ± 7.44, 48.7 ± 6 l / s, 3 ± 14, 44, 48%, 48.7 ± 6 l / s, 3 ± 14, 08 days, 34,51 ± 14,23, 25, and 0.89 in G1 compared to 4.26±2.98, 40.2±3 l/s, 4±66, 64%, 12 days, 46,11 ± 74.39, 38, and 1.88 times in G2.

4. DISCUSSION

COPD is a serious public health problem and a major cause of chronic morbidity and mortality in the world [10]. A study [11] found that the prevalence of COPD is 4.9% in Algeria. On the other hand, the incidence of bronchiectasis in Algeria is unknown; however, it is estimated that tuberculosis remains the greatest cause of bronchiectasis in the country.

The causes and mechanisms of bronchiectasis in COPD are still not clear.

One hundred twenty patients hospitalized for exacerbation of COPD were included in our study. These cases were divided into 2 groups: group G1 having 91 patients (not having associated bronchiectasis) and G2, including 21 patients with a combination of the two pathologies. The prevalence of this association was 18.75%. In our study, we included not only moderate to severe COPD cases but also mild cases, which could explain this prevalence. This prevalence was similar to the one found in a study conducted by M. Bafadhel [12] and was reported by Yong-hua Gao [13].

The increasing availability and the use of HRCT in COPD indicated that the association of this condition with bronchiectasis is frequent: *i.e.*, from 4 to 57% [14 - 16]. This prevalence varied according to regions and diagnostic criteria. In 71.42% of cases, bronchiectasis was located in the lower lobes, and it was cylindrical type in 80.95% of cases. In 76.19% of cases, it was bilateral. These characteristics were also reported by other studies. The basal seat, the cylindrical type, and the bilateral aspect of bronchiectasis were found, respectively, in 66.6%, 81.8%%, and 72.8% of cases in the study conducted by Eman O [17]. These results were also reported in other studies [18, 19]. The involvement of the lower lobes may reflect gravity-dependent retention of infected secretions [20].

Table 4. Multivariate logistic regression analysis for statistically significant variables (p<0.05).

	OR	95%C.I.	
FEV_1	4.3187	2.6301	6.8740
Purulent sputum	1.4150	0.5012	2.4189
Length of hospital stay	3.9617	0.8972	5.4633
SGQR	1.9442	0.3167	3.7410
CRP	0.9662	0.2187	1.8286
Exacerbations in previous year	0.7523	0.1437	1.5562

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the HSP70 proteins have the most sensitive response to changes in environmental temperature [7]. In cattle the relationship between gene expression and heat stress have been studied for the *HSP60*, *HSP70*, and *HSP90* genes in *Bos taurus indicus* [8-12] and *Bos taurus taurus* [5,13]. Despite to the progress of research in this topic, only one report in Simmental cattle has been published on the regulation of expression of *HSP70* gene but, none on the regulation of expression of *HSP60* and *HSP90* genes under heat stress.

The aim of this study was to determine levels of expression of *HSP60*, *HSP70*, and *HSP90* genes in Simmental cattle raised in tropical environments of Mexico.

MATERIALS AND METHODS

Locations

Between 2018 and 2019, Simmental cattle were sampled from three farms located in the tropical region of western Mexico. Samples were collected at Tangancícuaro in Michoacán State, Puerto Vallarta in Jalisco State, and Compostela in Nayarit State. Environmental temperature (T) and relative humidity (RH) were measured using weather stations installed in each farm. Temperature-humidity index (THI) was estimated with the following equation:

$$THI = [0.8 \times T\ (°C) + (RH\ [\%]/100) \times (T\ [°C] - 14.4) + 46.4]\ [14].$$

Certified veterinarians by the Mexican government obtained the samples. Owners of sampled individuals gave consent for the use of data for research purposes.

Procedures

One hundred and sixteen Simmental cattle were included in this study, of which, 60, 35, and 21 were from Tangancícuaro, Puerto Vallarta and Compostela, respectively. The ages of the animals ranged from nine months to six years (103 females and 13 males). The study was carried out from June to August of 2018 and 2019, when highest temperatures occurs in these States. Samples were collected in each animal twice a day, in the early morning from 04:00 to 06:00 h, when the temperature is lower, and at noon from 13:00 to 15:00 h, when the temperature is the highest of the day. Cows and calves grazed mostly on Llanero grass (Andropogon gayanus) and Mombaza grass (*Panicum maximum* cv. Mombasa) and were supplemented with a 2.5% molasses-urea mix (1 kg/head/d) during the months of March, April, and May. All animals had *ad libitum* access to minerals (40% salt; 56% calcium orthophosphate; 4% trace minerals) all year round.

A volume of 5 mL of blood were collected from the coccygeal vein in tubes with ethylenediaminetetraacetic acid that were kept cold until they arrived to the laboratory of the National Genetic Resources Center. In the laboratory, mRNA was obtained from whole blood using the commercial method SV Total RNA Isolation System (Promega, Madison, WI, USA). Later, mRNA concentration and purity were determined by spectrophotometry in a NanoDrop 2000 equipment (Thermo Fisher Scientific, Waltham, MA, USA). The synthesis of cDNA was carried out by reverse-transcribed of mRNA using GoScriptTM Reverse Transcription System (Promega, USA).

Gene expression analysis were carried out on *HSP60*, *HSP70*, and *HSP90* genes using the relative quantification by ΔΔCt comparative method. Constitutive *β-actin* was included as a reference gene. Target genes were amplified as duplex quantitative polymerase chain reaction (qPCR) in a StepOnePlus Real-Time System (Thermo Fisher Scientific, USA) with primers and marked probes shown in Table 1. The probes of target genes were marked with fluorophore FAM and probe of *β-actina* was marked with fluorophore HEX, while, quencher used in all probes was TAMRA.

Final volume of PCR reaction mix was 15 μL, which in-

Table 1. Description of primers and probes used in the amplification of *HSP60*, *HSP70*, and *HSP90* genes in Simmental cattle

Gene	Sequence of oligo (5' - 3')	Amplicon size (pb)	Alignment temperature (°C)
HSP60	F- GCGCTAAACTTGTTCAAGATGTTGC	159	65
	R- CTAACATCACACCTCTCCTGATTTC		
	P- CGGGGGATGGCACCACTACTGCT		
HSP70	F- CGAAAAACATGGCTATCGGCATC	135	62
	R- CTACGTGGCCTTCACCGATAC		
	P- GTTCCAGCACGGCAAGGTGGAGATC		
HSP90	F- TGCAGGAGGGTCCTTCACAG	102	65
	R- CTCCAGGTACTCAGTCTGGTC		
	P- CACAGGAGAACCAATGGGACGTGGAA		
β-actin	F- ACTCGTACGTGGGGGATGAG	94	62, 65
	R- TCCATGTCGTCCCAGTTGGTG		
	P- GAGAGGCATCCTGACCCTCAAGTAC		

HSP, heat shock proteins; F, forward primer; R, reverse primer; P, probe.

cluded 1X of TaqMan Fast Advanced Master Mix (Thermo Fisher Scientific, USA), 0.4 μM of each primer, 0.2 μM of probe and 0.2 μg of DNA. For the *HSP60* gene, the amplification conditions were 95°C for 10 min for pre-denaturation and 45 cycles at 95°C for 15 s, followed by 65°C for 1 min, while, for the *HSP70* and *HSP90* genes were 95°C for 10 min for pre-denaturation and 45 cycles at 95°C for 15 s, followed by 65 and 62°C, respectively for 40 s and 72°C for 40 s (Table 1).

Statistical analysis

Expression of *HSP60*, *HSP70*, and *HSP90* genes. The PROC MIXED procedure was used in a repeated measures analysis [15], considering that two samples were taken each day from the same animal (sampling hour). Hence, the mixed model approach to the analysis of repeated measures was used to model the appropriate covariance structure of data. Criteria for selecting the adequate covariance structure were Akaike [16] and Schwarz [17]. Factors included in the model were year (2018, 2019), farm (three locations), sex (males, females), sampling hour (am, pm), age of animals as a covariate, and the interaction of sampling hour with year, farm and sex. Comparison of means was carried out by lsmeans test based on the generalized least-squares method. The statistical analysis included animals with complete relative quantification in which the gene amplification was obtained for the *HSP* gene and the reference gene *β-actin*, and with Ct values lower than 40, otherwise data was discarded.

Physiological data. Body temperature (BT°), heart rate (HR), and respiratory rate (RR) were measured twice a day in each animal. Analysis of variance were carried out for BT°, HR, and RR including the fixed effects of sex and farm [15].

RESULTS AND DISCUSSION

Expression of *HSP60*, *HSP70*, and *HSP90* genes in Simmental cattle for sampling hour, year, farm sex and the interaction sampling hour×farm are shown in Table 2. For *HSP60* gene expression differences were found only for sex ($p = 0.0349$), in contrast, for the gene *HSP70* differences were observed for sampling hour ($p = 0.0042$), farm ($p<0.0001$), sex ($p = 0.0476$) and the interaction sampling hour×farm ($p = 0.0002$). For the gene *HSP90*, differences of expression were observed for farm ($p<0.0001$) and year ($p = 0.0521$) (Table 2).

In the analysis of sampling hour, between early morning (0.388±0.083) and noon (0.608±0.064), a significant difference ($p = 0.0042$) was observed for the *HSP70* gene expression which was lower in the morning than noon (Table 3). By contrast, Bretanha et al [5] reported in Simmental and Angus breeds that the expression of the *HSF1* and *HSP70* genes was similar in animals exposed to heat stress and control, in different sampling hours. In *HSP60* and *HSP90*, results found in this report are comparable to the obtained in Nerole and Caracu breeds, which, the expression of *HSP60* was similar during morning and afternoon in animals exposed in sun

Table 2. p-values for the expression of *HSP* genes for Simmental cattle in three farms of the tropical region of western México

Factor	*HSP60*	*HSP70*	*HSP90*
A: sampling hour	0.8220	0.0042	0.9848
B: year	0.1526	0.2955	0.0521
C: farm	0.2676	<0.0001	<0.0001
D: sex	0.0349	0.0476	0.3626
Interaction A×C	-	0.0002	-
Covariable: age	-	0.0651	-

HSP, heat shock proteins.

Table 3. Least-squares means and standard errors for the expression of *HSP* genes for Simmental cattle in three farms of the tropical region of western México

Factor	Level	*HSP60*	*HSP70*	*HSP90*
A: sampling hour	am	2.122±0.287[a]	0.388±0.083[a]	3.377±0.802[a]
	pm	2.067±0.287[a]	0.608±0.064[b]	3.392±0.802[a]
B: year	2018	1.807±0.253[a]	0.528±0.064[a]	4.651±0.549[a]
	2019	2.381±0.387[a]	0.468±0.075[a]	2.117±1.225[b]
C: farm	Compostela	1.915±0.417[a]	0.947±0.141[a]	5.138±0.984[a]
	Puerto Vallarta	1.850±0.352[a]	0.208±0.056[b]	0.116±0.857[b]
	Tangancícuaro	2.518±0.331[a]	0.340±0.059[c]	4.899±0.903[a]
D: sex	Female	1.553±0.219[a]	0.399±0.050[a]	2.891±0.663[a]
	Male	2.636±0.464[b]	0.597±0.102[b]	3.878±1.060[a]
Interaction A×C	am×Compostela	-	0.531±0.204[ab]	-
	pm×Compostela	-	1.363±0.141[c]	-
	am×Puerto Vallarta	-	0.258±0.074[ab]	-
	pm×Puerto Vallarta	-	0.157±0.056[a]	-
	am×Tangancícuaro	-	0.377±0.069[b]	-
	pm×Tangancícuaro	-	0.304±0.059[b]	-

HSP, heat shock proteins.
[a-c] Different letter superscripts in the same column indicate significant differences ($p \leq 0.05$).

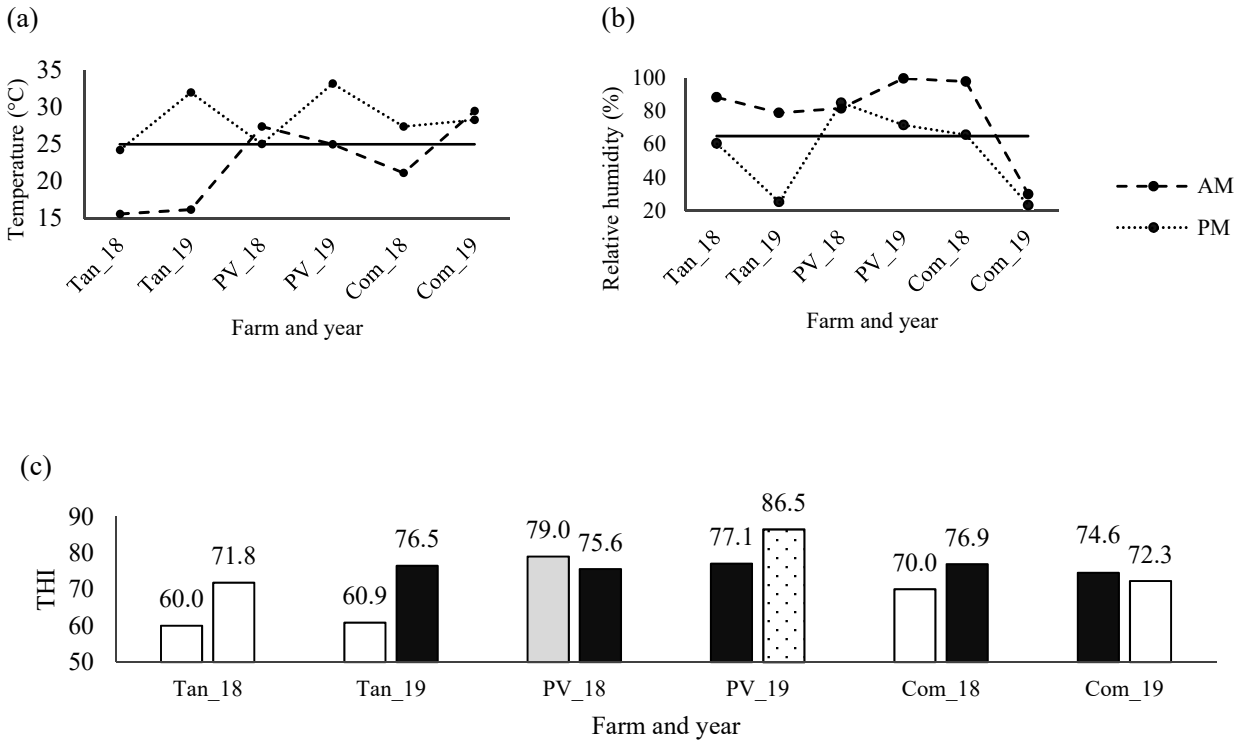


Figure 1. (a) Environmental temperature and (b) relative humidity of three sampled farms. Dashed lines depict to measurement in early morning; dotted lines depict to measurement in noon; solid line is temperature of 25°C in (a) and relative humidity of 65% in (b), respectively. (c) Temperature-humidity Index from three farms in 2018 and 2019, during early morning and noon. White bars show comfort conditions; black bars show heat stress alert; gray bar shows dangerous; dotted bar shows emergency. The first bar in each farm is THI value in early morning and the second bar is THI value at noon. Tan_18 = Tangancícuaro, 2018; Tan_19 = Tangancícuaro, 2019; PV_18 = Puerto Vallarta, 2018; PV_19 = Puerto Vallarta, 2019; Com_18 = Compostela, 2018; Com_19 = Compostela, 2019.

Table 4. Least-squares means and standard errors for farm and sex considering body temperature (BT°), heart rate (HR) and respiratory rate (RR) of Simmental cattle in the tropical region of western México

Items	BT°	HR	RR
Farm			
Tangancícuaro	38.83±0.092[a]	92.73±3.24	34.43±2.39[a]
Compostela	39.00±0.091[ab]	93.59±3.21	39.02±2.36[b]
Puerto Vallarta	39.32±0.080[b]	99.25±2.80	69.09±2.06[c]
Sex			
Males	39.18±0.06[a]	96.85±2.20	45.95±1.63[a]
Females	38.89±0.02[b]	94.90±0.86	42.35±0.63[b]

[a-c] Different letter superscripts in the same column indicate significant differences ($p \leq 0.05$).

Puerto Vallarta had the highest averages ($p<0.05$) for BT° and RR, which points out Puerto Vallarta as the place having the hottest weather, and the farm with animals under a great heat stress.

CONCLUSION

Expression of *HSP70* gene in Simmental herds of the tropical region of western México was different during early morning and noon, but the expression of the *HSP60* and *HSP90* genes was similar. *HSP70* gene showed to be a better marker of heat stress than *HSP60* and *HSP90* genes. Individual adaptability of the animals on exposure to heat stress could be influenced by *HSP* gene expression. Identification of resilient animals to heat stress would allow their incorporation in genetic improvement programs for the Simmental breed.

CONFLICT OF INTEREST

We certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

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Table 1. Univariate Descriptive Statistics of All Colon Surgeries and All Complex Surgical Site Infections (SSIs)

Variable	All Colon Surgeries (N=28,188)	Complex SSIs (N=720)	Non-complex SSIs (N=27,468)	*P* Value Comparing SSIs and Non-SSIs
Age, median y, (SD)	62 (14)	59 (15)	64 (14)	<.01[a]
Age >75 y, no. (%)	5,106 (18.1)	90 (12.5)	5,061 (18.3)	**<.01**[b]
Sex, male, no (%)	13,157 (46.7)	356 (49.4)	12,801 (46.6)	.14[b]
BMI (kg/m2, median, Q1–Q3)	27.8 (23.9–32.4)	28.2 (24.1–32.9)	27.8 (23.9–32.4)	.63[a]
Obesity (BMI > 30, %)	9,306 (33.0)	257 (35.7)	9,049 (32.9)	.13[b]
Operation time, min (Q1–Q3)	131 (92–187)	157 (113–218)	130 (90–186)	.07[a]
Operation time >75th percentile = 187 min (%)	6,935 (24.6)	257 (35.7)	463 (2.2)	**.01**[b]
Diabetes (%)	1,797 (7.3)	45 (7.2)	1,752 (7.3)	.81[b]
ASA 1–2 (low, %)	9,621 (34.1)	223 (31.0)	9,398 (34.2)	
ASA 3–5 (high, %)	18,070 (65.9)	497 (69.0)	18,070 (65.8)	.07[b]
Wound class (C, CC, %)	22,252 (78.9)	520 (72.2)	21,732 (79.1)	
Wound class (Co, D, %)	5,936 (21.1)	200 (27.8)	5,736 (20.9)	**<.01**[b]
Emergency (%)	2,774 (9.8)	120 (16.7)	2,654 (9.7)	**<.01**[b]
Open procedure (%)	15,951 (56.6)	450 (62.5)	15,501 (56.4)	**.01**[b]
Hospital volume >500 procedures	24,237 (86.0)	635 (88.2)	23,602 (85.9)	.09[b]

Note. BMI, body mass index; ASA, American Society of Anesthesiologists. Bold indicates statistical significance.
[a]Unpaired *t* test.
[b]χ^2 test.

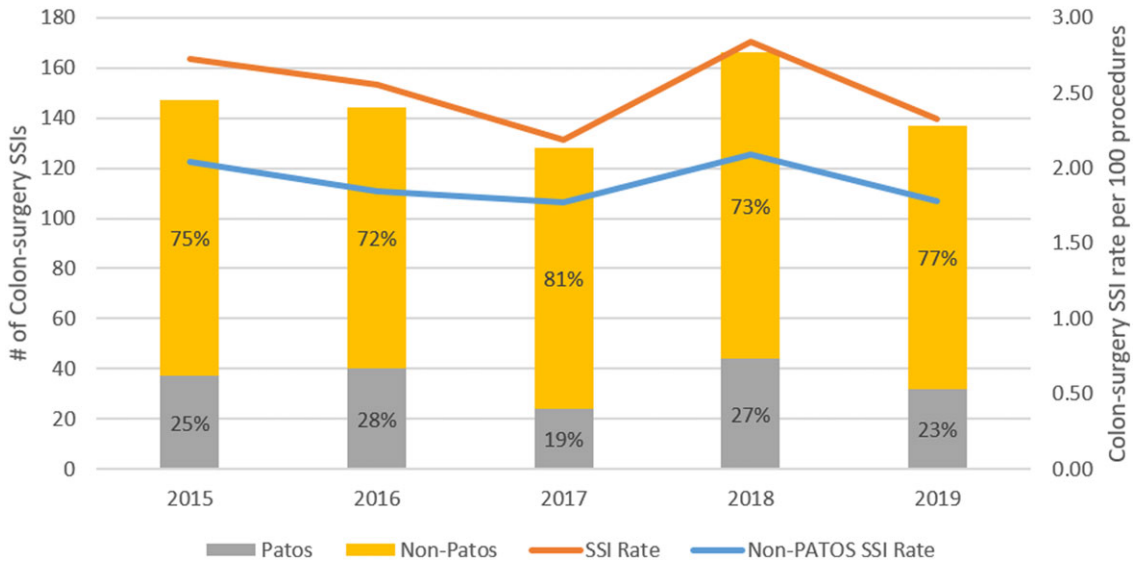


Fig. 1. Prevalence of complex colon surgery SSIs from 2013 to 2018, stratified by PATOS status.

(OR, 1.75; 95% CI, 1.50–2.04), contaminated or dirty wound class (OR, 1.27; 95% CI, 1.07–1.50), emergency procedure (OR, 1.87; 95% CI, 1.53–2.29), open surgery (OR, 1.20; 95% CI, 1.03–1.39), and hospital volume >500 procedures during the study period (OR, 1.35; 95% CI, 1.08–1.69) (Table 3, model 1).

In model 2, we compared patients with complex non-PATOS COLO SSIs to patients without complex non-PATOS COLO SSIs. Variables independently associated with complex non-PATOS COLO SSIs included age> 75 years (OR, 1.37; 95% CI, 1.06–1.75), operation duration >187 minutes (OR, 1.82; 95% CI, 1.53–2.16), and contaminated or dirty wound class (OR, 1.28; 95% CI, 1.03–1.60) (Table 3, model 2).

In model 3, we compared patients with and without complex PATOS COLO SSIs. Significant variables from this model included contaminated or dirty wound class (OR, 2.80; 95% CI, 2.12–3.71), emergency surgery (OR, 1.88; 95% CI, 1.45–2.44), and hospital volume >500 surgeries (OR, 2.02; 95% CI, 1.22–3.33) (Table 3, model 3).

Among the 200 complex COLO SSIs that were classified as having a contaminated or dirty wound class, 76 (38%) occurred after

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Table 2. Univariate Descriptive Statistics Comparing Complex PATOS COLO SSIs and Complex Non-PATOS COLO SSIs

Variable	Non-PATOS SSIs (N=544)	PATOS SSIs (N=176)	*P* Value Comparing PATOS SSIs and Non-PATOS SSIs
Age, median y (SD)	59 (16)	60 (15)	.46
Age >75 y, no. (%)	74 (13.6)	16 (9.1)	.15
Male	273 (50.2)	83 (47.2)	.49[b]
BMI, median kg/m^2 (Q1–Q3)	28.3 (24.1–33.1)	28.0 (24.2–32.6)	.72[a]
Obesity (BMI > 30), no. (%)	191 (35.1)	66 (37.5)	.59[b]
Operation time, min (Q1–Q3)	160 (116–222)	148 (104.5–202.5)	.08[a]
Operation time >75th percentile = 187 min (%)	203 (37.3)	54 (30.7)	.12[b]
Diabetes mellitus (%)	36 (8.0)	9 (5.1)	.23[b]
ASA 1–2 (low, %)	176 (32.4)	47 (26.7)	
ASA 3–5 (high, %)	368 (67.6)	129 (73.3)	.19[b]
Wound class (C, CC, %)	446 (82.0)	74 (42.1)	
Wound class (Co, D, %)	98 (18.0)	102 (58.0)	**<.01**[b]
Emergency (%)	59 (10.9)	61 (34.7)	**<.01**[b]
Open procedure (%)	328 (58.5)	132 (75.0)	**<.01**[b]
Hospital volume >500 procedures	471 (86.6)	164 (93.2)	**.02**[b]

Note. PATOS, present at the time of surgery; COLO, colon surgical procedure; SSI, surgical site infection; BMI, body mass index; ASA, American Society of Anesthesiologists. Bold indicates statistical significance.
[a]Unpaired *t* test.
[b]χ^2 test.

Table 3. Variable Analysis for Complex COLO SSIs Using Multivariable Logistic Regression Models[a]

Variable	Model 1 Odds Ratio (N=720)	*P* Value	Model 2 Odds Ratio (N=544)	*P* Value	Model 3 Odds Ratio (N=176)	*P* Value
Age >75 y (%)	**1.56 (1.23–1.96)**	**<.01**	**1.37 (1.06–1.75)**	**.01**	1.39 (0.93–2.13)	.10
Sex, male	1.08 (0.51–1.25)	.29	1.13 (0.96–1.33)	.15	1.00 (0.82–1.24)	.96
Obesity, BMI > 30 (%)	1.04 (0.89–1.21)	.61	1.02 (0.86–1.22)	.80	1.04 (0.85–1.28)	.69
Operation time >75th percentile = 187 min (%)	**1.75 (1.50–2.04)**	**<.01**	**1.82 (1.53–2.16)**	**<.01**	1.13 (0.85–1.52)	.40
Diabetes mellitus diagnosis (%)	0.93 (0.68–1.25)	.61	1.00 (0.71–1.40)	.99	0.85 (0.48–1.52)	.59
ASA score 3–5 (%)	1.17 (1.00–1.38)	.05	1.15 (0.96–1.39)	.13	0.94 (0.73–1.39)	.61
Dirty or contaminated wound class (%)	**1.27 (1.07–1.50)**	**.01**	**1.28 (1.03–1.60)**	**.03**	**2.80 (2.12–3.71)**	**<.01**
Emergency surgery (%)	**1.87 (1.53–2.29)**	**<.01**	1.28 (0.97–1.68)	.08	**1.88 (1.45–2.44)**	**<.01**
Open procedures (%)	**1.20 (1.03–1.39)**	**.02**	1.11 (0.93–1.32)	.23	1.30 (0.98–1.72)	.07
Hospital volume >500 procedures during study period	**1.35 (1.08–1.69)**	**.01**	1.05 (0.82–1.35)	.67	**2.02 (1.22–3.33)**	**.01**

Note. COLO, colon surgical procedure; SSI, surgical site infection; BMI, body mass index; ASA, American Society of Anesthesiologists; PATOS, present at the time of surgery. Bold indicates statistical significance.
[a]The outcome of model 1 was complex COLO SSIs. The outcome of model 2 was non-PATOS complex COLO SSIs. The outcome of model 3 was PATOS complex COLO SSIs.

emergent procedures and 102 (51%) were identified as PATOS. Of these 76 complex COLO SSIs that were emergent procedures and had a contaminated or dirty wound class, 51 (67%) were described as PATOS (Fig. 2). Conversely, of the 176 complex PATOS COLO SSIs, 61 (35%) were emergent procedures, 102 (58%) had contaminated or dirty wound class, and 51 (29%) were emergent and had contaminated or dirty wound class. Likelihood ratio tests found no significant effect measure modification between the emergent surgery and hospital volume variables ($P = .31$), laparoscopic surgery and hospital volume variables ($P = .41$), or contaminated or dirty wound and hospital volume variables ($P = .13$). However, a significant interaction was present between contaminated or dirty wound class and emergent surgery ($P < .01$).

Of the 2,774 emergent procedures in the data set, 2,042 (73.6%) were performed at high-volume centers. Furthermore, 2,076 (74.5%) of the emergent procedures were open procedures. Lastly, 4,838 (81.5%) of the contaminated or dirty wounds were seen at high-volume centers.

The distribution of pathogens among complex COLO SSIs consisted of typical enteric flora (Table 4). The proportion of specific organisms among PATOS COLO SSIs and non-PATOS colon SSIs was similar. However, the proportion of polymicrobial SSIs was

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Table 4. Comparison of Pathogens That Caused Complex Non-PATOS SSIs and Complex PATOS SSIs

Organism	Isolates From Complex SSIs, No. (%)	Isolates From Non-PATOS SSIs, No. (%)[a]	Isolates From PATOS SSIs, No. (%)[b]
Escherichia coli	257 (36)	187 (34)	60 (34)
Enterococcus spp	177 (25)	135 (25)	42 (24)
No pathogen identified	138 (19)	105 (19)	33 (19)
***Klebsiella* spp**	**63 (9)**	**57 (10)**	**6 (3)**
***Candida* spp**	**64 (9)**	**38 (7)**	**26 (15)**
***Bacteroides* spp**	**51 (7)**	**47 (9)**	**4 (2)**
Streptococcus spp	37 (5)	28 (5)	9 (5)
Staphylococcus aureus	33 (5)	27 (5)	6 (3)
Pseudomonas spp	28 (4)	21 (4)	7 (5)
Proteus spp	20 (3)	18 (3)	2 (1)
Enterobacter spp	20 (3)	16 (3)	4 (2)
Clostridium spp	19 (3)	11 (2)	8 (5)
Citrobacter spp	17 (2)	16 (3)	1 (1)
Coagulase-negative staphylococci	14 (2)	9 (2)	5 (3)
Polymicrobial infection	**437 (61)**	**378 (69)**	**59 (33)**

Note. PATOS, present at the time of surgery; SSI, surgical site infection.
[a]% based on 544 non-PATOS SSIs.
[b]% based on 176 PATOS SSIs.

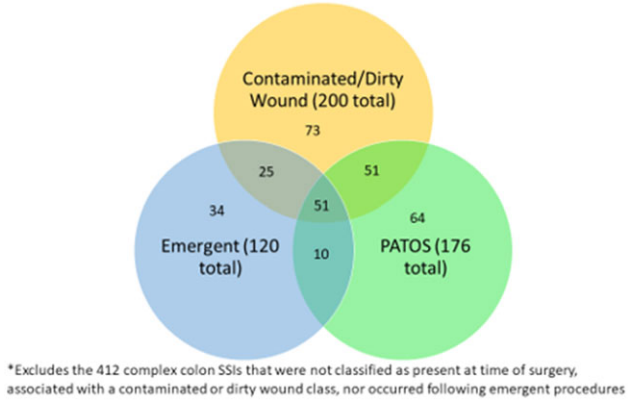


Fig. 2. Categorization of 308 complex colon SSIs by contaminated or dirty wound class, present at time of admission (PATOS), and emergent procedure.

higher among complex non-PATOS COLO SSIs compared to complex PATOS COLO SSIs (0.69 vs 0.33; $P < .01$).

Discussion

This 4.5-year, multicenter, retrospective cohort study is the largest to describe the impact of PATOS SSI on the epidemiology of colon SSIs. Removing PATOS eliminated ~25% of complex colon SSIs that would have previously been attributed to hospitals. Different variables were associated with PATOS and non-PATOS SSIs, suggesting that patients and procedures that lead to COLO SSIs characterized as PATOS have different characteristics than patients and procedures that lead to COLO SSIs not characterized as PATOS. In particular, emergency procedures, which are less amenable to quality improvement practices, were independently associated with PATOS SSI and were not associated with non-PATOS SSI. The results of this investigation support excluding PATOS SSIs from CMS reporting.

Our analysis identified several significant differences between procedures who developed PATOS versus non-PATOS COLO surgery SSIs. The COLO SSIs differed in surgical characteristics, including wound class, emergent procedure, endoscopic procedure, and surgical center volume. These differences are likely because emergent procedures were more commonly performed at high-volume centers (73.6%) and were open (74.5%) rather than laparoscopic procedures. Similarly, wounds that were categorized as contaminated or dirty were also more commonly seen at high-volume centers (81.5%). The EMM between contaminated or dirty wound class and emergent procedure was significant. These associations imply that patients who undergo colon procedures at high-volume hospitals are more likely to have features that are difficult to modify.

The definition of PATOS requires that evidence of infection be noted intraoperatively at the depth of subsequent SSI and documented within the narrative portion of the operative note. We found that only 102 (58%) of the PATOS COLO SSIs were associated with contaminated or dirty wounds. We expected that the categories of PATOS and contaminated or dirty wound class would have a larger amount of overlap. This finding raises questions about the accuracy of wound class data and/or application of the PATOS definition. At the very least, these data confirm that the PATOS definition continues to require review for specific documentation instead of inferring its presence based on wound class. To the best of our knowledge, this overlap has not been thoroughly evaluated in prior studies and may be the focus of a future prospective audit of wound class documentation.

Moreover, the non-PATOS COLO surgery SSIs and PATOS COLO surgery SSIs had different frequencies of polymicrobial

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multiple logistic regression. A p value of <0.05 was considered significant.

Results

Baseline characteristics of each group

Baseline characteristics are presented in Table 1. The proportion of patients aged >60 years was significantly higher in the OSCC group than in the control group (p = 0.019). Additionally, the prevalence of betel quid chewing, cigarette smoking, and alcohol consumption was significantly greater in the OSCC group than in the control group (all p < 0.001). Overall, the OSCC group had a higher tumor stage and more advanced tumor T status than did the control group (Table 1).

Table 1. The distributions of demographical characteristics in 1174 controls and 1254 male patients with oral squamous cell carcinoma.

Variable	Control group (N=1174)	OSCC group (N=1254)	p value
Age (years)			
< 60	761 (64.8%)	755 (60.2%)	0.019*
≥ 60	413 (35.2%)	499 (39.8%)	
Betel quid chewing			
No	978 (83.3%)	398 (31.7%)	
Yes	196 (16.7%)	856 (68.3%)	< 0.001*
Cigarette smoking			
No	552 (47.0%)	260 (20.7%)	
Yes	622 (53.0%)	994 (79.3%)	< 0.001*
Alcohol drinking			
No	942 (80.2%)	773 (61.6%)	
Yes	232 (19.8%)	481 (38.4%)	< 0.001*
Stage			
I+II		566 (45.1%)	
III+IV		688 (54.9%)	
Tumor T status			
T1+T2		604 (48.2%)	
T3+T4		650 (51.8%)	
Lymph node status			
N0		850 (67.8%)	
N1+N2+N3		404 (32.2%)	
Metastasis			
M0		1247 (99.4%)	
M1		7 (0.6%)	
Cell differentiation			
Well differentiated		204 (16.3%)	
Moderately or poorly differentiated		1050 (83.7%)	

N: number, OSCC: oral squamous cell carcinoma
Mann-Whitney U test was used between healthy controls and patients with oral cancer. * p value < 0.05 as statistically significant.

Genotype frequencies of RORB SNPs in each group

In the multiple logistic regression analysis, the distributions of all three *RORB* SNPs exhibited no significant differences between the OSCC and control groups (all p > 0.05; Table 2). However, the *RORB* SNP rs10781247 variants were significantly associated with moderate to poor cell differentiation in the OSCC group (AOR: 1.403, 95% CI: 1.011 to 1.947, p = 0.042; Table 3). Additionally, the *RORB* SNP rs10781247 variants were significantly associated with moderate to poor cell differentiation in the patients with OSCC who were betel quid chewers (AOR: 1.486, 95% CI: 1.025 to 2.155, p = 0.036), whereas no significant association was observed between *RORB* SNPs and clinical characteristics of patients with OSCC who were nonbetel quid chewers (all p > 0.05; Table 4). Moreover, *RORB* SNP rs3750420 variants were significantly associated with larger tumor size in individuals with buccal mucosa cancer (AOR: 1.542, 95% CI: 1.027 to 2.315, p = 0.036). However, no significant association was observed between *RORB* SNP rs3750420 genotypes and clinical characteristics of OSCC in the participants with tongue cancer (all p > 0.05; Table 5).

Table 2. Odds ratio and 95% confidence interval of oral squamous cell carcinoma associated with *RORB* genotypic frequencies.

Variable	Controls (N=1174) (%)	Patients (N=1254) (%)	AOR (95% CI)#	p value
rs3750420				
TT	380 (32.4%)	408 (32.5%)	1.000 (reference)	
TC	597 (50.9%)	617 (49.2%)	0.944 (0.765-1.166)	0.593
CC	197 (16.7%)	229 (18.3%)	0.995 (0.753-1.314)	0.970
TC+CC	794 (67.6%)	846 (67.5%)	0.957 (0.784-1.168)	0.665
rs10781247				
AA	312 (26.6%)	322 (25.7%)	1.000 (reference)	
AG	598 (50.9%)	650 (51.8%)	1.194 (0.953-1.497)	0.124
GG	264 (22.5%)	282 (22.5%)	1.148 (0.876-1.504)	0.317
AG+GG	862 (73.4%)	932 (74.3%)	1.180 (0.952-1.462)	0.130
rs17611535				
CC	1004 (85.5%)	1093 (87.2%)	1.000 (reference)	
CT	159 (13.5%)	156 (12.4%)	0.923 (0.699-1.219)	0.573
TT	11 (1.0%)	5 (0.4%)	0.361 (0.106-1.226)	0.102
CT+TT	170 (14.5%)	161 (12.8%)	0.884 (0.673-1.160)	0.372

AOR: adjusted odds ratio, CI: confidence intervals, N: number
The adjusted odds ratio with their 95% confidence intervals were estimated by multiple logistic regression models after controlling for age, betel quid chewing, cigarette smoking, and alcohol drinking.

Bioinformatics analysis of RORB expression in head and neck squamous cell carcinoma

Analysis of Cancer Genome Atlas data revealed that *RORB* mRNA levels were significantly higher in patients with head and neck squamous cell carcinoma than in those with noncancerous tissues (p = 0.0002; Figure 1A). Furthermore, *RORB* mRNA levels were significantly higher in the stage II tumors than in stage I tumors (p = 0.0463). Moreover, the *RORB* mRNA levels were significantly higher in stage IV tumors than in stage III tumors (p = 0.0252; Figure 1B).

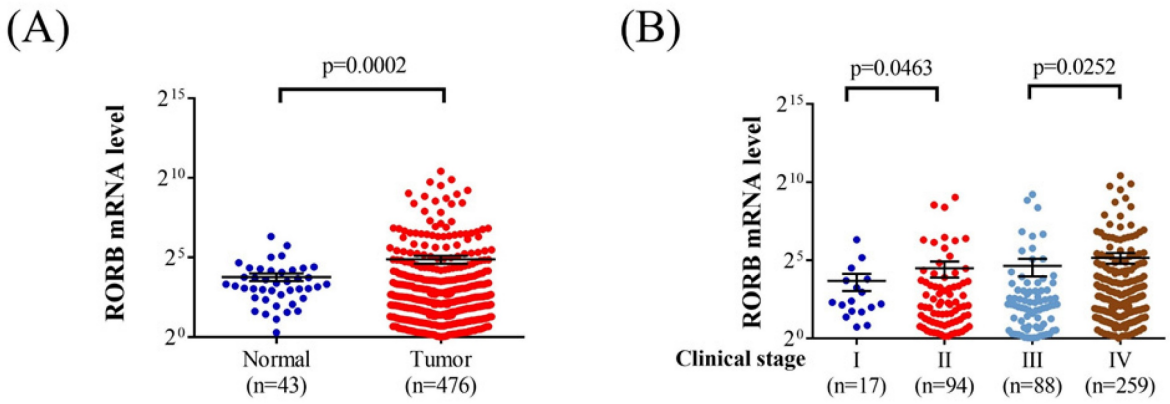


Figure 1. The Analysis of RORB mRNA by The Cancer Genome Atlas database. (A) The RORB mRNA level in head and neck squamous cell carcinoma patients and noncancerous tissues. (B) The RORB mRNA level in head and neck squamous cell carcinoma patients with different tumor stage.

Table 3. Odds ratio and 95% confidence intervals of clinical statuses associated with genotypic frequencies of *RORB* rs10781247 in male oral squamous cell carcinoma patients.

Variable	AA (N=322)	AG+GG (N=932)	AOR (95% CI)#	*p* value
Clinical Stage				
Stage I+II	154 (47.8%)	412 (44.2%)	1.000 (reference)	0.260
Stage III+IV	168 (52.2%)	520 (55.8%)	1.157 (0.897-1.492)	
Tumor size				
≦ T2	161 (50.0%)	443 (47.5%)	1.000 (reference)	0.445
> T2	161 (50.0%)	489 (52.5%)	1.104 (0.857-1.422)	
Lymph node metastasis				
No	221 (68.6%)	629 (67.5%)	1.000 (reference)	0.705
Yes	101 (31.4%)	303 (32.5%)	1.054 (0.803-1.384)	
Metastasis				
M0	321 (99.7%)	926 (99.4%)	1.000 (reference)	0.489
M1	1 (0.3%)	6 (0.6%)	2.080 (0.249-17.342)	
Cell differentiated grade				
Well	64 (19.9%)	140 (15.0%)	1.000 (reference)	0.042*
Moderate or poor	258 (80.1%)	792 (85.0%)	1.403 (1.011-1.947)	

AOR: adjusted odds ratio, CI: confidence intervals, N: number

The adjusted odds ratio with their 95% confidence intervals were estimated by logistic regression models.

* *p* value < 0.05 as statistically significant.

Table 4. Clinical statuses and genotypic frequencies of *RORB* rs10781247 in oral squamous cell carcinoma who are betel quid chewers or not betel quid chewers.

Variable	Betel Quid Chewers (N=856)				Non-Betel Quid Chewers (N=398)			
	AA (N=230)	AG+GG (N=626)	AOR (95% CI)#	*p* value	AA (N=92)	AG+GG (N=306)	AOR (95% CI)#	*p* value
Clinical Stage								
Stage I+II	110 (47.8%)	289 (46.2%)	1.000 (reference)	0.666	44 (47.8%)	123 (40.2%)	1.000 (reference)	0.193
Stage III+IV	120 (52.2%)	337 (53.8%)	1.069 (0.790-1.447)		48 (52.2%)	183 (59.8%)	1.364 (0.854-2.179)	
Tumor size								
≤ T2	111 (48.3%)	310 (49.5%)	1.000 (reference)	0.744	50 (54.3%)	133 (43.5%)	1.000 (reference)	0.066
> T2	119 (51.7%)	316 (50.5%)	0.951 (0.703-1.287)		42 (45.7%)	173 (56.5%)	1.549 (0.969-2.474)	
Lymph node metastasis								
No	159 (69.1%)	426 (68.1%)	1.000 (reference)	0.763	62 (67.4%)	203 (66.3%)	1.000 (reference)	0.851
Yes	71 (30.9%)	200 (31.9%)	1.051 (0.759-1.457)		30 (32.6%)	103 (33.7%)	1.049 (0.638-1.722)	
Metastasis								
M0	229 (99.6%)	621 (99.2%)	1.000 (reference)	0.572	92 (100.0%)	59 (99.7%)	1.000 (reference)	---
M1	1 (0.4%)	5 (0.8%)	1.844 (0.214-15.866)		0 (0.0%)	1 (0.3%)	---	
Cell differentiation								
Well	53 (23.0%)	105 (16.8%)	1.000 (reference)	0.036*	11 (12.0%)	35 (11.4%)	1.000 (reference)	0.891
Moderate or poor	177 (77.0%)	521 (83.2%)	1.486 (1.025-2.155)		81 (88.0%)	271 (88.6%)	1.051 (0.511-2.164)	

AOR: adjusted odds ratio, CI: confidence intervals, N: number

The adjusted odds ratio with their 95% confidence intervals were estimated by multiple logistic regression models after controlling for cigarette smoking, and alcohol drinking.

* p value < 0.05 as statistically significant.

Table 5. Clinical statuses and genotypic frequencies of *RORB* rs3750420 in buccal mucosa cancer and tongue cancer

Variable	Buccal mucosa cancer (N=433)				Tongue cancer (N=409)			
	TT (N=140)	TC+CC (N=293)	AOR (95% CI)#	p value	TT (N=130)	TC+CC (N=279)	AOR (95% CI)#	p value
Clinical Stage								
Stage I+II	69 (49.3%)	133 (45.4%)	1.000 (reference)	0.448	62 (47.7%)	126 (45.2%)	1.000 (reference)	0.632
Stage III+IV	71 (50.7%)	160 (54.6%)	1.169 (0.781-1.750)		68 (52.3%)	153 (54.8%)	1.107 (0.729-1.680)	
Tumor size								
≤ T2	81 (57.9%)	138 (51.8%)	1.000 (reference)	0.036*	66 (50.8%)	141 (50.5%)	1.000 (reference)	0.965
> T2	59 (42.1%)	155 (52.9%)	1.542 (1.027-2.315)		64 (49.2%)	138 (49.5%)	1.009 (0.666-1.530)	
Lymph node metastasis								
No	98 (70.0%)	205 (70.0%)	1.000 (reference)	0.994	79 (60.8%)	174 (62.4%)	1.000 (reference)	0.757
Yes	42 (30.0%)	88 (30.0%)	1.002 (0.645-1.554)		51 (39.2%)	105 (37.6%)	0.935 (0.610-1.433)	
Metastasis								
M0	139 (99.3%)	292 (99.7%)	1.000 (reference)	0.592	130 (100.0%)	277 (99.3%)	1.000 (reference)	---
M1	1 (0.7%)	1 (0.3%)	0.476 (0.030-7.667)		0 (0.0%)	2 (0.7%)	---	
Cell differentiation								
Well	22 (15.7%)	55 (18.8%)	1.000 (reference)	0.436	17 (13.1%)	33 (11.8%)	1.000 (reference)	0.720
Moderate or poor	118 (84.3%)	238 (81.2%)	0.807 (0.469-1.386)		113 (86.9%)	246 (88.2%)	1.121 (0.600-2.097)	

AOR: adjusted odds ratio, CI: confidence intervals, N: number

The adjusted odds ratio with their 95% confidence intervals were estimated by multiple logistic regression models after controlling for cigarette smoking, and alcohol drinking.

* p value < 0.05 as statistically significant.

Discussion

In the present study, variants of the *RORB* SNP rs10781247 were associated with moderate to poor cell differentiation in both the overall OSCC group and the subgroup of patients with OSCC with a history of betel quid chewing. Moreover, variants of the *RORB* SNP rs3750420 were associated with a higher tumor T status in patients with buccal mucosa cancer. Furthermore, *RORB* mRNA levels were significantly higher in patients with head and neck squamous cell carcinoma, with the highest expression levels observed in those with advanced tumor stages.

The *RORB* gene has been implicated in various diseases [14, 15]. The *RAR-related orphan receptor* (*ROR*) family regulates the circadian rhythm and influences sleep patterns, with sleep duration affected by *RORA* and its SNP rs75981965 [23]. Additionally, the *ROR* family plays a role in the aging process, where both *RORA* and *RORB* independently contribute to the risk of cognitive aging [14]. In addition to circadian regulation, the *ROR* family is involved in the growth of specific cell types, such as epithelial cells and osteocytes [24], and research has demonstrated that *RORA* expression is suppressed during the development of OSCC [18]. The expression of *RORB* is also correlated with prognosis in breast cancer [25], with lower levels observed in the patients with endometrial cancer [24]. Zheng *et al.*, reported that *RORB* expression was significantly associated with changes in the tumor immune microenvironment in head and neck squamous cell carcinoma [16]. Beyond gene expression, the genetic polymorphisms of *RORB* can influence disease development. For instance, the *RORB* SNP rs7867494 is associated with an increased incidence of breast cancer [17], and multiple *RORB* SNP variants are associated with the development of lung and prostate cancers [26]. In addition to the *RORB* gene family, several other genetic factors contribute to the development of OSCC [13, 27, 28]. The presence of *CDKN2A* was found to be significantly correlated with OSCC development [29], and mutations in the *p53* gene are commonly observed in patients with OSCC [27]. Genetic polymorphisms, such as the *interleukin-10* SNP-A592C, are more frequently observed in individuals with OSCC than in wild-type carriers [30], and the SNP rs1412115 A>G on chromosome 10 was demonstrated to increase the risk of OSCC in a Chinese population [31]. Additionally, *GAS5* SNP rs145204276 (Ins/Del or Del/Del) variants are associated with a higher risk of moderate to poor cell differentiation in oral cancer [32]. Because both *RORB* and its SNPs have been demonstrated to regulate cancer growth [17, 24] and because *RORA* levels are significantly higher in individuals with OSCC [18], *RORB* SNP variants may be corrected with the clinical status of OSCC. This hypothesis is supported by the results of the present study.

In the present study, the distribution of *RORB* SNP rs10781247 variants was more commonly found in patients with OSCC with moderate to poor cell differentiation. Another study also reported that *RORB* SNP rs10781247 variants were more common in individuals with cognitive aging [14]. The association between *RORB* SNP rs10781247 variants

128 kD and 116 kD, vacuolating toxin A (VacA) with bands at 95 kD and 91 kD, urease B (ureB) with a band at 66 kD, and urease A (ureA) with a band at 30 kD. Type I (virulent/carrying the CagA gene) *H. pylori* infection was determined by the presence of CagA and/or VacA bands, type II (non-virulent/not carrying the CagA gene) *H. pylori* infection was indicated by the presence of urease antibodies only, without CagA and VacA bands, and negative results for all three antibodies were classified as negative for *H. pylori* antibodies. The experiments were conducted strictly following the instructions, using reagents purchased from Shenzhen Brault Biological Company with production batch number 161216.

Statistical analysis

The data were analyzed using SPSS 19.0 statistical software. Count data were presented as frequency and percentage and compared using the χ^2 test. Measurement data were expressed as mean ± standard deviation (x ± s) and compared using the t-test. Logistic regression analysis was conducted to identify influencing factors. Statistical significance was determined at a value of p of less than 0.05. The data were statistically analyzed using SPSS 19.0 statistical software. Enumeration data were presented as the number of cases and percentages and were compared using the χ^2 test. Measurement data were presented as mean ± standard deviation (x ± s) and were compared using the *t*-test. Logistic regression analysis was used to identify influencing factors. A *p*-value less than 0.05 was considered statistically significant.

Results

Analysis of the status of *H. pylori* infection in patients with scrub typhus

Among the 117 patients with scrub typhus, 105 tested positive for *H. pylori*, resulting in an infection rate of 89.7% (105/117). This included 79 cases (67.5%) of type I *H. pylori* infection and 26 cases (22.2%) of type II *H. pylori* infection. Among the 130 healthy subjects, 71 cases were positive for *H. pylori*, resulting in an infection rate of 54.6% (71/130). This included 39 cases (30.0%) of type I *H. pylori* infection and 32 cases (24.6%) of type II *H. pylori* infection. The prevalence of *H. pylori* infection and type I *H. pylori* infection was higher in patients with *tsutsugamushi* disease compared to healthy individuals ($\chi^2 = 6.225$, $p = 0.013$; $\chi^2 = 12.314$, $p < 0.001$), respectively. The difference was not statistically significant when comparing the prevalence of type II *H. pylori* infection ($\chi^2 = 0.122$, $p = 0.727$). The details were showed in Table 2.

Multifactorial regression analysis of the relationship between different types of *H. pylori* infection and scrub typhus

Multiple regression analysis as used to further explore the relationship between different types of *H. pylori* infection and scrub typhus. After adjusting for gender, age and place of residence, the results revealed that type I *H. pylori* infection was an independent risk factor for the occurrence of scrub typhus (2.407; 1.249–4.64; 0.009). The details were showed in Table 3.

The correlation between different types of *H. pylori* infection and the presence of multiple organ damage in patients with scrub typhus

Among the 117 patients diagnosed with scrub typhus, 45 individuals (38.5%) were found to have more than three comorbidities,

TABLE 2 *H. pylori* infection status in patients with *tsutsugamushi* disease.

	Scrub typhus patients (*n* = 117)	Healthy subjects (*n* = 130)
H. pylori, *n* (%)	105 (89.7)	71 (54.6)
Type I, *n* (%)	79 (67.5)	39 (30.0)
Type II, *n* (%)	26 (22.2)	32 (24.6)

TABLE 1 Comparison of general characteristics between *H. pylori* positive and *H. pylori* negative patients with Scrub typhus disease.

Characteristics	Scrub typhus patients (*n* = 117)		χ^2/*t*-value	*p*-value
	H. pylori (+) (*n* = 105)	*H. pylori* (–) (*n* = 12)		
Male, [*n* (%)]	45 (42.9)	6 (50.0)	0.084	0.771
Age (±s), y	56.1 ± 14.89	61.25 ± 15.99	1.981	0.262
Residency, [*n* (%)]			6.026	0.014
Rural	65 (61.9)	3 (25.0)		
Urban	40 (38.1)	9 (75.0)		
complication [*n* (%)]			3.808	0.051
≥ 3 kinds of complications	44 (41.9%)	1 (8.3%)		
< 3 kinds of complications	61 (58.1%)	11 (91.7%)		
H. pylori [*n* (%)]				
Type I	79 (75.2)	0 (0)		
Type II	26 (24.8)	0 (0)		

TABLE 3 Multivariate regression analysis of the relationship between *H. pylori* infection and *tsutsugamushi* disease.

Variable	β coefficient	Standard error	Wald χ^2	*p*-value	OR (95% Confidence interval)
Sex (male/female)	−0.087	0.319	0.075	0.785	0.916 (0.49,1.714)
Residency (Rural/Urban)	−0.208	0.332	0.393	0.531	0.812 (0.423,1.557)
Age (≥65/<65)	0.158	0.355	0.199	0.656	1.172 (0.584,2.351)
H. pylori (type I/type II)	0.878	0.335	6.88	0.009	2.407 (1.249,4.64)

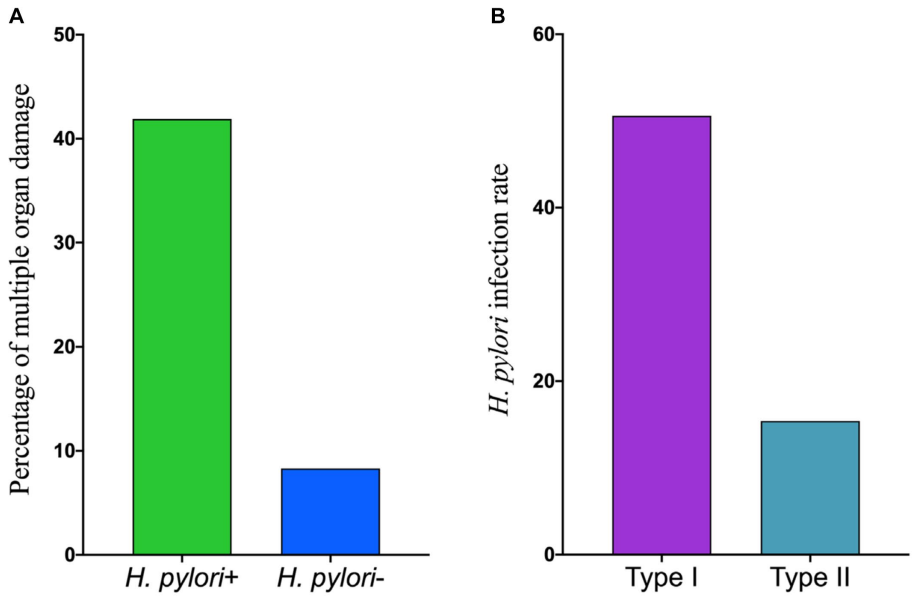


FIGURE 1
The correlation between different types of *H. pylori* infection and the presence of mutiple organ damage in scrub typhus patients. **(A)** The percentage of mutiple organ damage in scrub typhus patients. **(B)** The infection rate of different *H. pylori* type.

which were defined as multiple organ damage. Further analysis was conducted to examine the different subtypes of *H. pylori* infection among patients with scrub typhus who experienced multiple organ damage. Out of the 45 patients suffering from scrub typhus with multiple organ damage, the prevalence of *H. pylori* positivity was observed to be 41.9% (44/105), whereas the *H. pylori* negativity accounted for 8.3% (1/12). These proportions did not exhibit any statistically significant differences ($x^2 = 1.858$, $p = 0.173$). The proportion of *H. pylori* infections with type I was 50.6% (40/79), which was significantly higher than that with type II *H. pylori* infection rate of 15.4% (4/26), ($x^2 = 4.735$, $p = 0.030$). For detailed information, please refer to Figure 1.

Discussion

Helicobacter pylori infection is commonly acquired during childhood and if left untreated, it can persist for a lifetime (Frenck and Clemens, 2003). In contrast, scrub typhus is an acute infectious disease caused by opportunistic infection of the host with *Orientia tsutsugamushi*. The status of *H. pylori* infection in scrub typhus patients and whether *H. pylori* infection has an impact on the occurrence and development of scrub typhus has not been reported. Epidemiological surveys have shown that the infection rate of *H. pylori* infection is related to social economic situation, living conditions, sanitary conditions and living habits, occupation and quality of drinking water (Xu Wenting and Chengfu, 2022). The prevalence of scrub typhus is also found in specific populations (e.g., military personnel or plantation workers) and agricultural staff in townships (Devamani et al., 2022). To avoid any influence of environmental and occupational factors on the study results, we strictly matched the regional distribution, age, and gender of the *Tsutsugamushi* patients and the healthy medical examination population.

This study has made a novel discovery, demonstrating that the prevalence of *H. pylori* infection was significantly higher in scrub typhus patients compared to healthy subjects (89.7% vs. 53.8%, $p < 0.05$). This suggests that individuals infected with *Orientia tsutsugamushi* are more likely to develop typical scrub typhus if they already have an existing *H. pylori* infection. This finding may help explain why not all *Orientia tsutsugamushi* infected hosts develop

result in adverse respiratory effects, but a growing body of research is showing the links between PM2.5 and cardiovascular health (Orr et al., 2020; Landguth et al., 2020). Globally, PM 2.5 exposure contributes to approximately 4 million deaths annually making it the fifth ranked risk factor for premature death (Thurston, 2018).

One of the most important aspects of the REACH program is its emphasis on local air quality research as a low-cost way to engage students in science practices – doing science. Students learn about timely and important environmental and health science content in parallel with learning science practices. Because the focus is on local air quality, the REACH program can be applied worldwide in both rural and urban settings and is easily tailored to addressing local air quality issues. The program is also easily integrated into the local schools' science or other curriculum. The relatively low-cost air sampling equipment needed is provided by the REACH program, and there are varying ways for teachers to implement the program within existing classes, so the entry costs are low for schools with limited resources. This adaptability makes it very easy to integrate the three NGSS dimensions within the REACH program, and within the classroom.

The REACH program and its predecessors have been described in previous manuscripts (Delaloye et al., 2016; Delaloye et al., 2018; Ward et al., 2016). There is a typical schedule of events for students who participate in the REACH program. At the start of the program, students learn background information about air pollution and impacts to health which often involves a member of the REACH team making an in-person or virtual presentation. Students are then organized into small two to four person groups and given guidance on conducting research. The student research groups then develop their own research projects investigating indoor air quality within their homes, schools, or other community settings to explore a variety of relevant and real-world science research questions related to indoor air quality and its impact on health. Via their teachers, students are provided with low-cost PM2.5 air sampling equipment and training so they can complete the data collection portion of their project. With guidance, students then process and analyze their data. Finally, towards the end of the school year, students practice science communication by presenting their research findings to others, including their peers. This takes place in a variety of ways, including local community events, culminating events such as the annual REACH symposia at the University of Montana campus, in class or school presentations to community members, at scientific conferences, or at other physical or virtual venues.

Research aimed at supporting the ongoing development of the program and evaluating the efficacy of early versions of the program has used mixed-method approaches based on surveys that include both fixed-response and free-response questions for students (Delaloye et al., 2016; Delaloye et al., 2018; Ward et al., 2016). Previous reports of quantitative findings from fixed-response questions provided evidence that the program may positively impact student self-efficacy for research skills as well as their performance on tests of science reasoning and experimental design; more than a third of students reported that their participation made them more interested in science as a content area and almost a quarter reported that participation increased their interest in a science career (Delaloye et al., 2018; Ward et al., 2016). "In addition, an overwhelming percentage of students (85%) rated their experience in the symposium component of the program as either 'good' (59%) or 'excellent' (26%) (Ward et al., 2016).

Unlike structured multiple choice or Likert-scaled questions, free-response questions can provide evaluators with additional contextual understanding of such quantitative findings along with unanticipated student-centered information that was not targeted by the fixed-response questions. Thematic analysis of student writing in response to free-response questions may result in the emergence of themes that provide additional insight into student perspectives and intended or unintended effects the program may have on participating students. Student responses to free-response questions over a three to four-year period are the focus of this manuscript.

METHODOLOGY

Surveys were administered in person at the end-of-year symposium on the University of Montana campus. Data was collected at symposia between the years of 2014 and 2018. At these symposia, high school students (primarily juniors and seniors) presented their air quality research projects in either a poster or oral presentation format to their peers and a panel of judges. The surveys were administered on paper and was composed of approximately 35 items including both fixed-response and free-response questions. The hand-written responses were converted to text in an Excel spreadsheet. Table 1 lists four free-response questions that are the focus of this manuscript.

Table 1. *Student survey free-response questions and years used in survey.*

Survey Question[a]	Years Used
What were the most important things you learned as a result of preparing for your presentation or poster this past school year?	2014, 2015, 2016, and 2018
What were the most important things you learned as a result of giving your presentation or poster today?	
Please explain how the program did or did not change your interest in science.	2015, 2016, and 2018
Please explain how the program did or did not change your interest in a science career.	

[a]*These questions were not used in the 2017 version of the survey.*

Using a process designed by the program's external evaluator, a three-person panel of REACH personnel used thematic analysis (Braun and Clarke, 2006; Braun et al., 2019) to identify patterns across the responses. The panel read the responses to each question a total of three times. Initially, each reviewer read the recorded responses separately while independently identifying themes in the responses. The reviewers then convened and developed consensus on a framework for coding the main common themes in the responses to each question. During the second reading, again done separately, each reviewer assigned these thematic codes to statements in each response. A single statement could be coded with more than one theme if different parts of the comment could be assigned to different themes. Finally, the reviewers reconvened to compare their individual coding and resolve any discrepancies on how they had assigned codes.

RESULTS

The first two survey questions analyzed were included in the 2014, 2015, 2016, and 2018 surveys. These questions asked the students to comment about what they learned from preparing their poster or presentation and what they learned from presenting their poster or presentation. A total of 426 individual student surveys were recorded. From those students a total of 337 comments about what was learned from preparing their poster or presentation were analyzed resulting in 425 classifications into eight themes. Another 318 comments about what was learned from presenting the research were analyzed resulting in 430 classifications into the same eight themes. Table 2 displays the themes that emerged from the student responses, along with example quotes from student comments and the percentage of student responses to these questions that were coded using each theme.

Each theme includes at least 3% of student responses that were categorized. Student comments about learning preparation and time management and learning effective speaking and presenting skills represented 28% and 17% of the categorized comments. Over one-fifth (21%) of student preparation responses and 5% of the presentation responses described learning about environmental health science research related to air quality or air pollution as a highlight of their participation in the program. More than half of student comments on presenting (52%) described gaining effective

Table 2. *Themes in student comments about what they learned from preparing and presenting summaries of their air quality research projects.*[b]

Theme	Description	Examples	Distribution of coded comments % (n)
1) Design and conducting an experiment.	Students stated that they learned how to design, conduct, and explain an experiment or science research project.	*How to clearly describe an experiment and its importance on the real world.* *It is important to have good data and to be able to accurately interpret this data.*	Preparing 11.0 (47) Presenting 3.0 (13)
2) Preparation and time management.	Students stated they learned that preparation and time management were instrumental in the success of their project.	*Time management and communication is extremely important over long periods of time.* *A project that spans over a year goes by faster than you think.*	Preparing 27.6 (117) Presenting 12.8 (55)
3) Working with a group.	Students stated that working with and coordinating with a group of fellow students was important to the success of their project.	*What it's like to work with a group when it's a challenge to get together.* *Working with your group collaboratively will help you succeed.*	Preparing 7.9 (34) Presenting 2.6 (11)
4) Air Quality knowledge and related information.	Students stated they gained new knowledge and appreciation of air quality and related environmental health science issues.	*Air quality is important and I did not realize how relevant it was.* *Particulate matter is and can be a problem for health.*	Preparing 21.7 (92) Presenting 5.2 (22)
5) Nature of science.	Students stated they learned something about the nature of science as an evolving process.	*You can't assume science. Experimenting is key.* *I learned that in all sciences conclusions lead to more questions.*	Preparing 3.2 (14) Presenting 0.9 (4)
6) Challenging, positive experience.	Students stated that the project was a challenging yet positive and rewarding experience.	*Data gathering and sharing isn't as easy as it seems.* *To be proud of the time and effort I put into my project.* *If you don't try, you won't learn.*	Preparing 8.3 (35) Presenting 22.8 (98)
7) New Skill: Effective speaking and construction of poster or presentation (science communication).	Students stated they learned a new skill that involved effective speaking about or presenting their project including how to relax/not stress.	*I learned that I could stand and be able to speak about something I had no knowledge on previously.* *How to portray my ideas to other people in a clean and organized way.*	Preparing 17.5 (74) Presenting 52.2 (225)
8) New Skill: Data management and/or analysis.	Students stated they learned a new skill that involved efficient and effective data management and analysis with the use of spreadsheets and graphing software. (Excel, Sheets)	*How to manage data/change data to graphs.* *We learned how to statistically analyze data.*	Preparing 2.8 (12) Presenting 0.4 (2)

[b]*N = 425 student comment classifications about preparing their presentation and 430 comment classifications about giving their presentation.*

speaking and presenting skills as the most important lesson learned from giving their research presentation, while 17% highlighted this same lesson learned from preparation. Almost a quarter (23%) described the benefit of giving their presentation in general terms as a challenging, positive experience. Approximately 8% of the comments indicated the preparation was a challenging, positive experience.

The second group of free-response questions from the survey related to how the REACH program affected their interest in science and their interest in a science career. These questions were part of the 2015, 2016, and 2018 surveys. A total of 341 student surveys were recorded and analyzed, with 364 comments classified based on science interest and another 375 comments classified based on science career interest. Table 3 displays the eight themes that were identified to represent patterns in the student responses, as well as select student comments from the surveys.

Comments about how the program changed the students' interest in science and interest in science careers indicated they valued the research experience (19% and 15.5%), val-

Table 3. *Themes in student comments about how the program affected their interest in science and science careers.*[c]

Theme	Description	Examples	Distribution of coded comments % (n)
1) Valued the research experience.	Students' comments indicated an appreciation of the research experience they gained in the program.	*It made my love of science great again because it let me have a hands-on experience in science.* *It made me think for myself rather than reading it out of a book.*	Science interest 19.0 (69) Science career interest 15.5 (58)
2) Valued the new knowledge and/or information.	Students' comments indicated an appreciation of new knowledge and /or information they gained during the course of the program.	*The program made me realize how much PM2.5 can affect you, and all sorts of normal everyday things can harm you… made (me) want to learn more.*	Science interest 15.9 (58) Science career interest 7.7 (29)
3) Valued the application of science to solve problems.	Students' comments indicated an appreciation of the application of science to solve problems or improve quality of life.	*It made me think about what makes our air quality bad and gave me an idea on how to fix it.* *It helped me to realize that no matter what field I base my career in I can make a difference.* *Being a part of something that will benefit future generations was a privilege.*	Science interest 7.7 (28) Science career interest 5.0 (19)
4) Valued the new perspective.	Students' comments indicated an appreciation of the new perspective about science they had gained from the program.	*It changed my interest in science because it showed a lot more aspects that can be studied.* *It was really interesting and informed me on a lot of things I had never thought of before.*	Science interest 3.6 (13) Science career interest 4.3 (16)
5) Reinforced interest in science or science career.	Students' comments indicated they had been interested in science or a science career prior to participating in the program.	*I already knew that I really liked science, wasn't much to change on my opinion.* *I was already interested in science before, the project only solidified this thinking.*	Science interest 11.0 (40) Science career interest 13.6 (51)
6) Interested in science or science career, but not air quality.	Students' comments indicated they had been interested in science or a science career prior to participating in the program, but they were not interested in air quality science.	*It did not greatly affect my outlook in science because it isn't really related to the field that I am interested in.* *I did not really enjoy this project, so as long as the science I do in the future is on other topics that I like better, then I'll be good.* *It expanded my knowledge on radon but I do not think I will be very interested in this field of science.*	Science interest 12.9 (47) Science career interest 19.7 (74)
7) Not interested in science or science career.	Students' comments indicated they were not interested in science or a science career prior to or after participating in the program.	*I don't like science at all so this program didn't help or wouldn't.* *This program didn't change my interest because science is never a subject I would like to study in the future.* *I have never really been interested in being a scientist.*	Science interest 10.4 (38) Science career interest 14.7 (55)
8) No reason given or other.	Students' comments were not classifiable in one of the above themes or there was no comment provided.	*It was a project and I needed the grade to pass.* *Would have preferred to spend more time on other things, not the project.*	Science interest 19.5 (71) Science career interest 19.5 (73)

[c]*N = 364 student comment classifications related to* ***science interest*** *and 375 comment classifications related to* ***science career interest.***

Table 1 Trial characteristics

Trial name	Population	Host trial intervention and comparison	Design of the host trial	Design of SWAT
GHT2000 [21]	Inactive 18–74-year-olds with hypertension, suspected hypertension, pre-hypertension or high-normal blood pressure	Interventions: (i) GP gym-based referral plus web tool, (ii) sport referral (iii) sport referral plus web tool Comparator: GP gym-based referral	Individually randomised four-arm group trial	Two-arm trial, individually randomised
PSM COPD [22]	COPD patients aged 18 years or older with mild dyspnoea	Intervention: a telephone-based self-management intervention Comparator: usual care	Individually randomised two-arm group trial	Two-arm trial, clustered by general practice
HI-Light [23]	Patients aged 5 years and over with vitiligo	Interventions: (i) Handheld narrowband UVB (NB-UVB) and (ii) a combination of potent topical corticosteroid and NB-UVB, compared with potent topical corticosteroid	Individually randomised three-arm, placebo-controlled	Two-arm trial, individually randomised
seAFOod [24]	Patients aged 55–73 years identified during screening colonoscopy as being at 'high risk' for subsequent surveillance colonoscopy	Interventions: (i) 2 g eicosapentaenoic acid per day and (ii) 300 mg aspirin per day Comparator: placebo	Individually randomised 2 by 2 factorial trial	Two-arm trial, cluster randomised (Endoscopy Unit taking part in the UK NHS Bowel Cancer Screening Programme)
STOP ACEi [20]	Patients with advanced progressive CKD receiving angiotensin-converting enzyme inhibitor or angiotensin receptor blockers (or both)	Intervention: Discontinue Angiotensin Converting enzyme inhibitor (ACEi)/Angiotensin Receptor Blocker (ARB) or combination of both Comparator: Continue ACEi, ARB or combination of both	Individually randomised two-arm group trial	Two-arm trial, individually randomised

a second group allocating patients on reminder letters after they had initially been contacted.

Five host trials provided data on recruitment [21–24]. Access to multimedia resulted in little or no difference in recruitment rates (pooled odds ratio = 0.96, 95% CI: 0.79 to 1.17 p-value = 0.671, $I^2 = 0\%$) (Table 2 and Fig. 1).

Four host trials provided data on participant acceptance rates [21–23]. Access to multimedia resulted in little or no difference in the likelihood of responding positively to the invitation compared to participants receiving standard information (pooled odds ratio 0.98, 95% CI: 0.85 to 1.13, p value = 0.778, $I^2 = 0\%$) (Fig. 2 and Table 3).

Three SWATs provided data on retention [22–24]. Access to multimedia resulted in little or no difference in retention compared to participants receiving standard information (pooled odds ratio 1.07, 95% CI: 0.71 to 1.62, p value = 0.737, $I^2 = 44.7\%$) (Fig. 3 and Table 4).

There was no marked difference in intervention effects by baseline rate (odds ratio 0.94 95% CI 0.65 to 1.35 in low baseline trials compared with 0.99 95% CI 0.73 to 1.33 in high baseline trials).

Table 2 Primary outcome—randomised to host trial

Study	Standard	Multimedia	Odds ratio (95% CI)	% weight
GHT2000	64/1049 (5.9%)	57/1048 (5.4%)	0.89 (0.61 to 1.28)	27.7
GHT2000 (reminder)	41/1057 (3.9%)	35/1055 (3.3%)	0.85 (0.54 to 1.35)	18.2
PSM COPD	247/2280 (10.8%)	185/1934 (9.6%)	0.84 (0.58 to 1.22)	28.2
HI-Light	51/1136 (4.5%)	54/1094 (4.9%)	1.11 (0.60 to 2.02)	10.4
seAFOod	61/395 (15.4%)	68/333 (20.4%)	1.44 (0.88 to 2.37)	15.5
Pooled	464/5917 (7.8%)	399/5464 (7.3%)	0.96 (0.79 to 1.17)	100.0

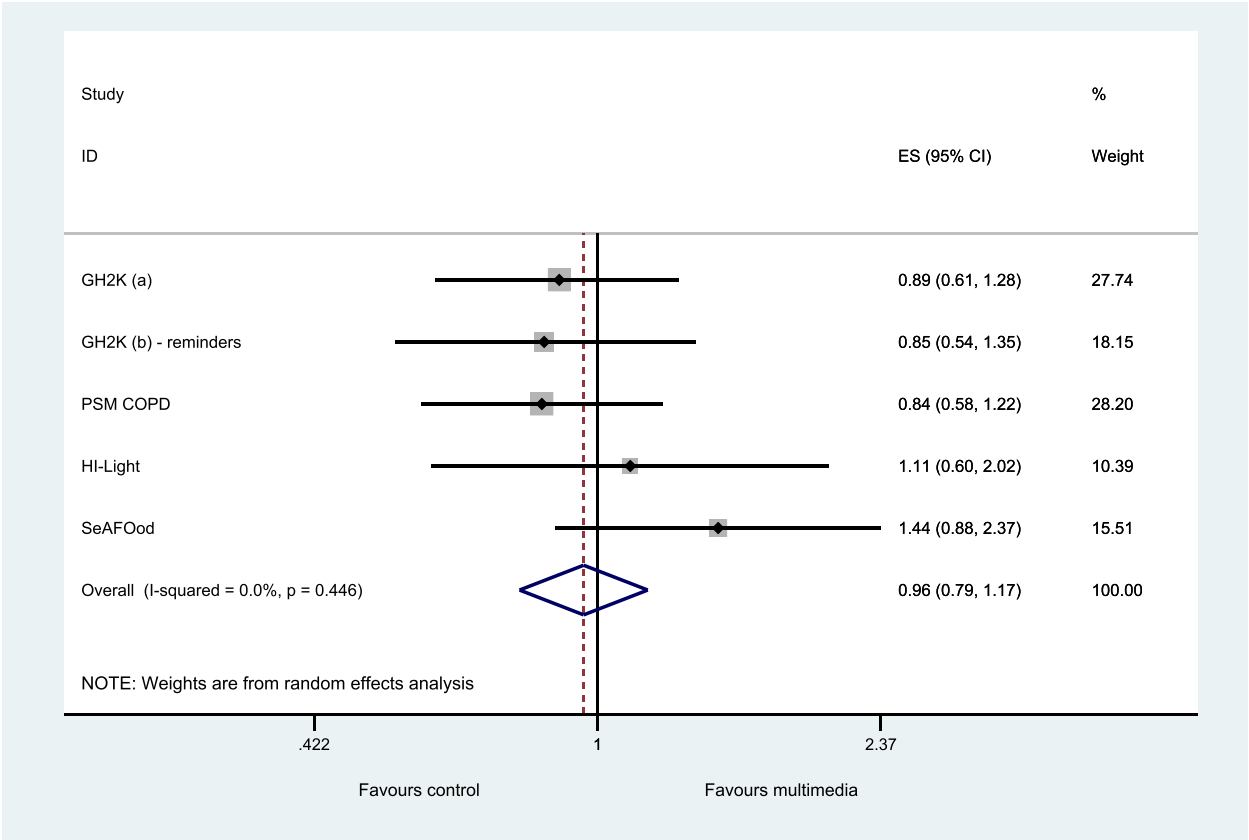


Fig. 1 Primary outcome—randomised to host trial. Heterogeneity chi-squared = 3.72, $p = 0.446$; $I^2 = 0.0\%$; test of pooled odds ratio = 1: $z = 0.42$, $p = 0.671$

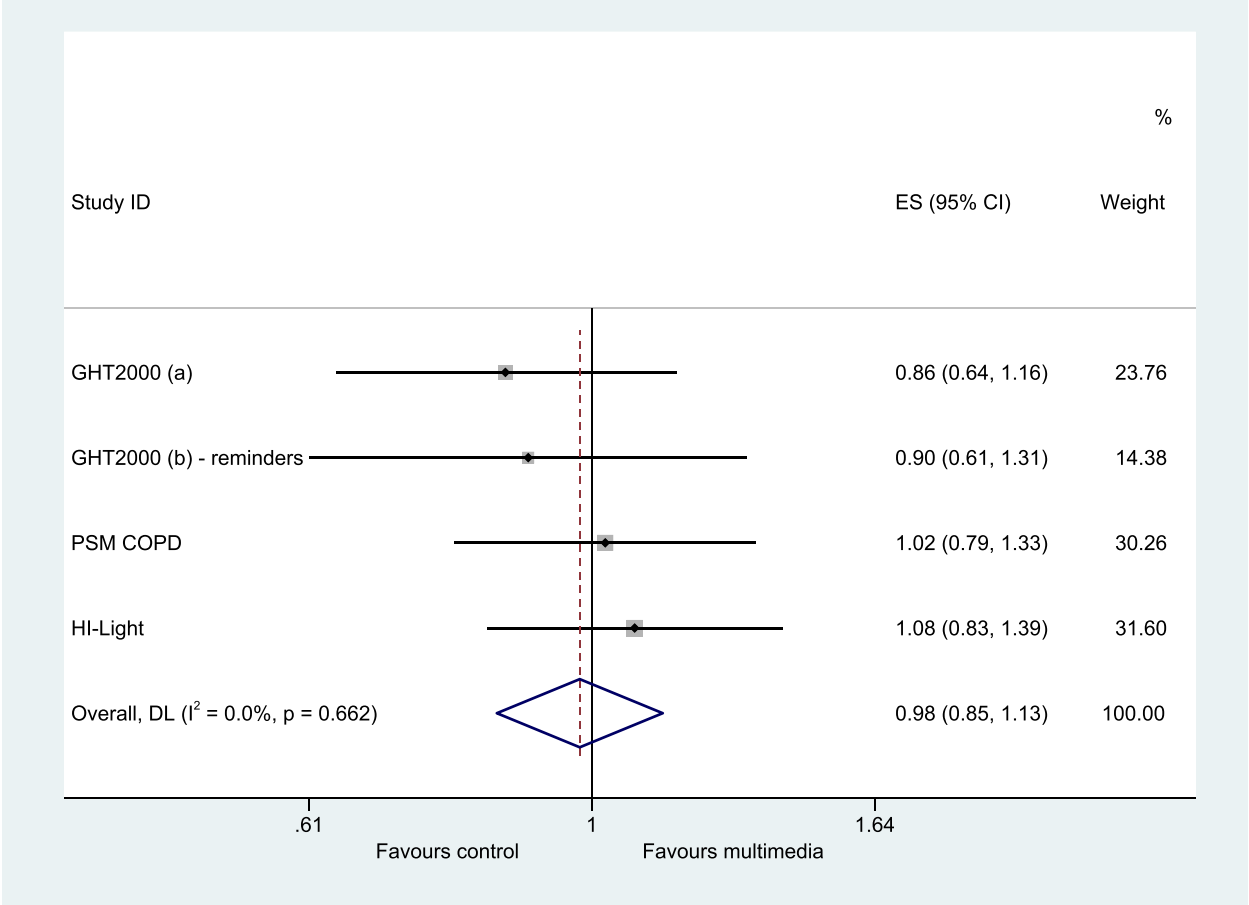


Fig. 2 Secondary outcome—responded positively to invitation. Heterogeneity chi-squared = 1.59, $p = 0.662$; $I^2 = 0.0\%$, test of pooled odds ratio = 1: $z = 0.28$, $p = 0.778$

Table 3 Secondary outcome—responded positively to invitation

Study	Standard	Multimedia	Odds ratio (95% CI)	% weight
GHT2000	100/1049 (9.5%)	87/1048 (8.3%)	0.86 (0.64 to 1.16)	23.8
GHT2000 (reminder)	59/1057 (5.6%)	53/1055 (5.0%)	0.90 (0.61 to 1.31)	14.4
PSM COPD	464/2280 (20.3%)	412/1934 (21.3%)	1.02 (0.79 to 1.33)	30.3
HI-Light	221/1136 (19.5%)	226/1094 (20.7%)	1.08 (0.83 to 1.40)	31.6
Pooled	844/5522 (15.3%)	778/5131 (15.2%)	0.98 (0.85 to 1.13)	100.0

Discussion

Summary

We tested the effects of access to multimedia information on trial recruitment and retention. In a number of SWAT evaluations that run through a diverse group of host trials, the intervention did not improve acceptance, recruitment or retention rates among participants.

Strengths and limitations of the study

This programme of coordinated SWATs was one of the first to be initiated, demonstrating the broad feasibility of this strategy as a model for the more rapid development of an evidence base.

As with most SWATs, there was no formal sample size calculation for individual trials. The host trials undertaking the SWATs were self-selected and therefore the

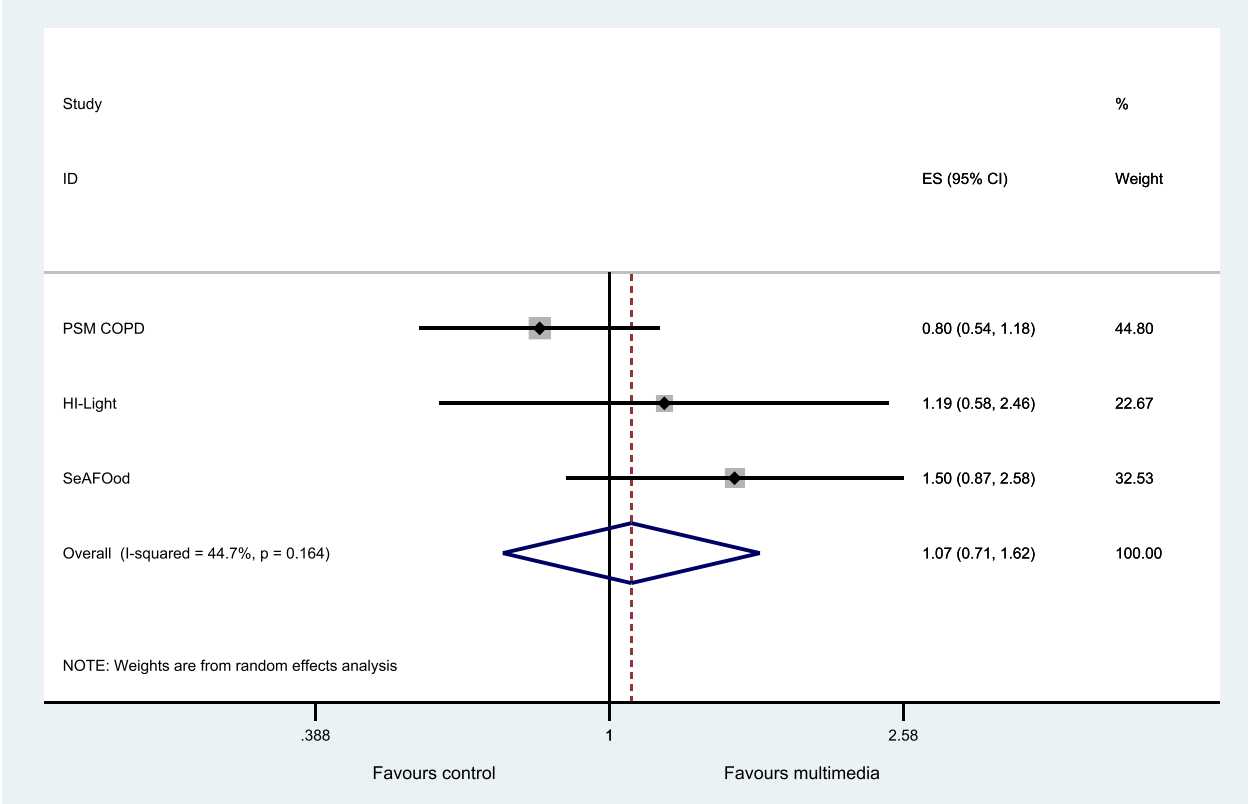


Fig. 3 Secondary outcome—retained at primary endpoint of the host trial. Heterogeneity chi-squared = 3.62, $p = 0.164$; $I^2 = 44.7\%$, test of pooled odds ratio = 1: $z = 0.34$, $p = 0.737$

Table 4 Secondary outcome—retained at primary endpoint of the host trial

Study	Standard	Multimedia	Odds ratio (95% CI)	% weight
PSM COPD (at 12 months)	223/2280 (9.8%)	159/1934 (8.2%)	0.80 (0.54 to 1.18)	44.8
HI-Light (at 9 months)	35/1136 (3.1%)	40/1094 (3.7%)	1.19 (0.58 to 2.46)	22.7
seAFOod (at 12 months)	49/395 (12.4%)	57/333 (17.1%)	1.50 (0.87 to 2.6)	32.5
Pooled	307/3811 (8.1%)	256/3361 (7.6%)	1.07 (0.71 to 1.62)	

studies on which the programme was run represent a relatively specific (though diverse) group of study contexts, with most patients in the older age groups. It is possible that the variation in those contexts was sufficient to give the recruitment strategy a fair test across multiple designs and populations, and there was limited evidence of significant variation in effect. Even the pooled analysis of data from six trials left some imprecision in the estimate of effect. We have provided broad details of the patient populations sought for each of the trials, but we do not have detailed information on the demographics of those who took part in the SWATs. It is important to note that the participants in the SWATs are not the same as those taking part in the trials, as many people participate in the SWATs without entering the trial. Collecting such information is complicated and potentially burdensome for trial teams. One study was unable to provide the necessary data for our main analysis, which reduced the sample size available. In our experience, problems with delivering SWATs are fairly rare but these difficulties do highlight that SWATs can stretch the resources of already busy trial teams.

Although we planned to assess the use of the multimedia intervention, including its various elements, to provide better context to our outcome data, an error in the web-hosting software meant no accurate data on use

PK drafted the paper. All authors read the manuscript, revised it critically, and approved the final manuscript.

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Availability of data and materials
The summary meta-analysis data is available from the corresponding author.

Declarations

Ethics approval and consent to participate
The START programme of recruitment strategies was approved by the National Research Ethics Service (NRES) Committee, Yorkshire and the Humber – South Yorkshire (Ref: 11/YH/0271) on 5 August 2011. Host trials had their own approvals.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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via WhatsApp to maintain confidentiality. All personally identifiable information was anonymised during transcription, and WhatsApp was encrypted to ensure data privacy; transcripts were stored on SharePoint in a password-protected computer.

Results

A total of 14 interviews were conducted; 12 were males, while 2 were females (Table 1). All participants occupied the deputy director level or above. Of the 14 participants, 10 were national level representatives from MoH, universities, and implementing partners (labelled as 'MoH') and 4 were representatives from funding and global organisations (labelled as 'PO'). Among all participants, six had more than 8 years; four had 5 to 7 years; and 4 had 2 years to 4 years of working experience leading digital health in their respective organisations. All five regions of the African continent were represented. We provide the participant number, age, and organization (01/44/MoH) for each participant quote.

In the following sections, we report findings under the three objectives.

Objective 1: Digital health maturity assessment contribution to strengthening health systems

Digital health systems maturity assessments were identified as a critical beginning step to strengthening health systems. Several countries had not conducted DHSMA. Health leaders cited maturity assessments contributions including systems performance review, prioritisation of resources, needs and strategies.

TABLE 1: Participants' profile.

Institution	Total	National level	International level
Ministry of Health	7	7	0
Universities	2	2	0
Implementing partners	1	1	0
Funding partner	2	0	2
Global organisation	2	0	2
Total	**14**	**10**	**4**

All participants agreed that there is a huge gap in conducting DHSMAs in their countries as a national approach, with only 3 out of 10 countries attesting to having conducted a digital health maturity assessment in the past 3–5 years. Similarly, one participant stated that strategic planning for DHSMAs has neither been performed nor prioritised at the continent level. Some participants confirmed being involved in implementing DHSMAs as a once-off, research-based or nationwide activity. Respondents cited that most of these initiatives were donor-driven; hence, they lacked follow-up to recommendations and ownership of results and did not prioritise countries' specific needs.

Participants indicated that a DHSMA contributes to knowing the system's performance and understanding the gaps and strengths to build a baseline to strengthen digital health. Other participants acknowledged that maturity assessments provide a platform to respond to systematic challenges and recommendations on data quality seen through data quality assurance (DQAs) and monitoring and evaluation (M&Es), which most organisations support.

Summary of objective 1

Table 2 summarises the contribution of digital health maturity assessments to strengthening HIS and provides examples of corresponding best practices from evidence cited by participants.

Objective 2: Facilitators and barriers to digital health maturity assessments

Governance and leadership

Government support and political will were identified as the main facilitators for promoting digital health maturity assessment. Policies and best practices backed by the MoH's digital health system priorities were considered as effective motivators for implementation. Policies reportedly empower health leaders by creating an enabling environment and allowing resource allocation to build a strong base for assessing digital health:

TABLE 2: Summary of digital health maturity assessments contribution to health information systems strengthening.

Summary point	Quote	Best practice cited by key informants
Identifies focus areas that need to be addressed in a digital health system, its strengths and prioritise action.	'If you don't know the existing systems' capacity and maturity, how can you make decisions and allocate resources to support this? So, the maturity assessment is critical in providing leaders with the information they need to plan and support efficient and optimal digital health strengthening.' (Participant 05, 39 years, MOH)	Zambia conducted the ISHO maturity assessments to strengthen the performance of the SmartCare digital health system.
Provides data for evidence-based decisions, including policies and guidelines that prioritise interventions to strengthen the digital health system.	'After we did the assessment, where we found the most weaknesses is where we focused. But I'm not necessarily saying that the other areas were not focused on. Still, we were able to focus on that area and have a number of initiatives listed in this area of workforce get to the level that we would want to be as a country.' (Participant 08, 36 years, MOH)	Kenya developed an online human resource capacitation platform, training on digital health systems as part of yearly accreditation.
Provides better advocacy opportunities for political will and prioritisation of resource allocation towards interventions to strengthen a digital health system.	'… [*W*]e have that's what we call ANICiiS, which is the entity, but before that had to do all kinds of assessment in the health sector to see what the gaps are, what needs to be done, which resulted in what we call the digital plan for the country.' (Participant 12, 42 years, MOH)	The Democratic Republic of Congo's national Digital Health System budget is 80% funded by the government's post-digital health maturity assessments.
Tracks longitudinal evolution and progress of digital health system performance and in providing accurate data	'… [*A*]fter the previous assessment, we then realized there are a number of issues that we needed to add looking at infrastructure, governance, the establishment of a health informatics and data analytics department, which is one critical area and addresses some issues pertaining to governance.' (Participant 03, 46 years, MOH)	Ethiopia conducted an ISHO assessment in 2020 and followed up with another assessment in 2023 to track the progress of its digital health system roadmap.
Digital health maturity assessments provide an opportunity to work on systematic challenges seen in DQAs and M&E, which most programme budget allocations prioritise	'Currently, we are planning another level of assessment because there's some level of inconsistency with HIV data in Nigeria. So, we didn't want to determine the exact number of people who are currently on treatment because our spectrum data was showing that Nigeria has just reached maturity level on treatment, yet we are still having new patients on treatment.' (Participant 01, 44 years, MOH)	Nigeria is planning an ISHO assessment to validate the performance of the digital health systems.

DQA, data quality assurance; M&E, monitoring and evaluation; ISHO, informatics-savvy health organisation; MoH, Ministry of Health; KIIs, key informants.

Objective 3: Ways health leaders can better plan for digital health maturity assessments

Participants provided various ways health leaders can better plan digital health maturity assessments in their countries.

All participants recommended multi-stakeholder collaborative engagement when planning maturity assessments, from the idea's conception to completion. Several participants attested that engagement is crucial to ensure that all parties understand and agree on the objectives and methodology of the assessment, leverage existing M&E systems and piggyback on already established systems, such as the DHIS2 monitoring, to effectively plan for a digital health maturity assessment. One participant emphasised that:

> 'First of all, the methodology must be understood in a consultative process supported by widespread stakeholder engagement early on in the process including both public and private sector within the health information area. The second part will be after that widespread consultative process, I would want the objectives for the National Health Information System maturity assessment to be outlined and linked to the strategic plan of the digital health system for all implementers' involvement.' (Participant 10, 46 years, PO)

In support of the aforesaid, several participants agreed that for digital health maturity assessment planning to be effective, there needs to be strong collaborative engagement of all stakeholders to advocate for government prioritisation and political will to support the initiative.

Several participants stated that collaborators should be involved in planning a maturity assessment based on their context. Another participant opined that stakeholders hold different powers and expertise, which is key when planning a maturity assessment for the country. Participants outlined the need to bring a sense of recognition, ownership and support to plan a participatory action-based maturity assessment. More than half of the participants supported planning for action by prioritising using context-based assessment tools, minimising duplication of activities, and letting countries decide the processes. In support, one participant echoed that:

> 'Stakeholders should not be restricted only to MoH's HIS and digital health department but include all partners, implementing, funding, Ministry of Information and the regulatory bodies, telecommunications [*public and private*], power supply organizations, community representatives and advocates. Including such key stakeholders necessitates them to understand the need for prioritizing digital health assessments and their role in setting goals and ownership of recommendations.' (Participant 14, 41 years, MoH)

Almost all (n = 11/14) participants highly prioritised increasing the use of data when planning for digital health maturity assessment. Respondents indicated that the available systems' data should be used to show the system's weaknesses or strengths for the leaders to focus the assessment. A few also cited that stakeholders should have access to and the ability to analyse or report the data, which would prepare them to understand assessment findings and take ownership of improving the system.

Summary of objective 3

Table 4 summarises recurring recommendations from participants.

Discussion

Health leaders recognised the value of having evidence from DHSMAs to guide them in planning for HIS strengthening. They recognised that DHSMAs provide a critical first step to enhance a system's performance status, highlighting areas to integrate, expand and scale up towards the long-term goal of a functional, optimised, sustained and strengthened health system. Barriers to implementing maturity assessments include a lack of skilled workforce knowledgeable in digital health maturity assessments, fragmented digital health systems using expensive infrastructure and a lack of

TABLE 3: Summary of barriers and facilitators to planning digital health maturity assessments.

HSS pillar	Barriers	Facilitators	Examples of digital health strategic documents
Leadership and governance	Lack of collaborators' buy-in, shared vision and supporting policies.	Coordinated implementation of digital health policies through a national HIS ministry	Ministry of Public Health, DRC. National Development Plan Health Informatics 2020–2024.[19]
Financing	There is no specific national digital health system budget to support digital health maturity assessments.	Allocate a specific national digital/HIS strengthening budget	Democratic Republic of The Congo Ministry of Public Health Quantified Roadmap of Digital Health Investments.[20]
Information systems	Fragmented systems that rely on power and connectivity.	Open-Source Systems that are interoperable and standardised building from external collaborations	Ministry of Public Health, DRC. National Development Plan Health Informatics 2020–2024.[19]
Workforce	Inadequate technically capacitated staff to properly understand, use, maintain and manage digital health systems infrastructure and technology.	Curriculum for digital health systems workforce and health workers orientation	Kenya's Virtual Academy online training for health workers.[21] Zimbabwe and Ethiopia informatics training within the University of Zimbabwe and the University of Gondar
Service delivery	Competing priorities Siloed and fragmented digital health systems divorced from the national HIS structure make choosing or integrating systems to focus maturity assessment difficult.	Collaborating with other organisations and agencies, leveraging partnerships to provide resources, expertise and support for digital health strengthening efforts	DRC's One Health digital health system Zambia's SmartCare System
Medical supplies	A lack of standard/user-friendly tools costly equipment and infrastructure, including maintaining server rooms, providing reliable connectivity at all levels.	Flexible resource allocation systems	Integration of mobile technology to supplement HIS

HIS, health information systems; HSS, health systems strengthening.

TABLE 4: Summary of recommendations of ways health leaders can better plan for digital health systems maturity assessments.

Recommendations when planning a maturity assessment	Key quote	Priority
Collaborative engagement through stakeholder mapping	'A multi-stakeholder, a multi-organization, multidisciplinary approach to the assessment, so ensuring that there's a lot of good participation from a key set of diverse stakeholders …' (Participant 06, 45 years, PO)	High
Identify the country's priorities and choose appropriate implementation tools, implementers, time and process.	'And finally, relevance to the local context, so, for the assessment to work well, it should be relevant to that local context so that findings apply to those specific needs of each country or organization. The data collection tool itself should be flexible to meet those needs.' (Participant 07, 45 years, PO)	High
Increase utilisation of available health data to grow the demand and value of digital health and have its assessment prioritised.	'I think we need to be very specific in the way that we solve the problems by using tools, for instance, these same digital health maturity tools, break them down even to the lowest level and keep ourselves alive all the time, like the way that people report monthly, or weekly, to keep health information systems alive or used through reporting and assessments …' (Participant 13, 51 years, PO)	High
Capacity building at all levels to ensure scalability and continuity of the assessment process	'One is that country leaders need capacity building and more sensitization. Some of these health leaders don't understand the digital health terminology, and the leaders need to be sensitized to the health information systems used to give more support.' (Participant 03, 46 years, MoH)	High
Identify DQA and M&E planned and include a digital health assessment to provide a cost-effective approach to conducting assessments that can be implemented at routine intervals.	'… [*A*]lso, being flexible in the data collection and in the assessment tool itself and applying scalability to already existing plans like M&E and DQA conducted by partners can be cost-effective and promote continuity of digital health maturity assessment, to address different types of systems sizes and complexities and having a cost-effective approach to conducting assessments …' (Participant 11, 43 years, MoH)	Medium
Advocate for increased political will, government ownership, inclusion in strategic plans and investment into digital health maturity assessments	'I feel like the best recommendation I would have is having the political will for and understanding the need for accurate data, then it is easy to mobilize for resources to ensure that maturity assessments are planned and happening because there is some form of accountability.' (Participant 12, 42 years, MoH)	Low

DQA, data quality assurance; M&E, monitoring and evaluation.

financing. Key facilitators to implementing digital health maturity assessment included coordination, collaborating with existing M&E programmes and knowledgeable health workers to conduct digital health assessments at all health facilities. Addressing these barriers and facilitators is crucial for effective digital health strengthening and data-driven decision-making in healthcare systems.

Governance is critical to HSS; maturity assessments, especially participatory assessments, can help strengthen that. Political will in LMICs can be a stepping stone to developing digital health policies that strengthen DHSMA to not only rely on international aid. Weak health systems governance in LMICs has resulted in fragmented or ad hoc health policy formulation, poising challenges in implementing DHSMA and impacting efforts to strengthen overall health systems.[22,23] Leadership and governance for digital health include having multisectoral digital health steering committees such as an eHealth Technical Working group that oversees the implementation of digital health, interoperability activities and financial resourcing to aid the implementation of recommendations from the assessment.[14,24,25] Information derived from maturity assessments can benefit digital health governance in: (1) lobby for political will to finance digital health systems, (2) providing guidance for improvement in health systems' policies and (3) improving efficiency, effectiveness, performance and productivity in the whole health system.[26] Through participatory planning, health systems governance leadership in Ghana and Rwanda effectively prioritised areas to improve digital health, supported by strong political will and governance structures.[27,28] The LMICs governments should start growing focus on making the DHSMAs sustainable for a health system to function optimally.

Growing knowledge about the value of a digital health maturity assessment is critical to their sustainable implementation. The over-reliance on donor funding has led to digital systems being heavily supported by international organisations in many LMICs, affecting long-term implementation and sustainability of digital health systems, as objectives shift to those of funders. Firstly, making DHSMAs routine and operationalised as part of a strategic vision could increase demand for sustainable digital health assessments. Secondly, efforts should focus on building the culture of conducting regular DHSMAs using integrated and decentralised approaches. Integration supports participatory planning to co-develop the goals/objectives of the assessment based on country needs, priorities and collaborators' implementation efforts.[29] Participatory planning addresses not only technical aspects but also cultural, structural and governance-related factors that contribute to an effective maturity assessment. Thirdly, adapting existing digital health assessment tools such as ISHO and Stages of Continuous Improvement (SOCI) to the country context could aid in routinising DHSMAs without high costs.[16,30,31] Fourthly, strengthening the capacity of health leadership in planning and conducting systems performance monitoring at all health facilities promotes accountability to health data, increasing data use. When the health team understands their responsibility, accountability and teamwork are cultivated, essential to improving data use, quality for informed decision-making, policy change and planning.[5,25] All these factors could potentially sustain implementation and maturity assessments using available resources without relying much on donor funds.

Although there is limited literature on barriers and facilitators to implementing DHSMAs, our findings resonate to some recommendations where maturity assessments have been conducted outside Africa. Provider experiences, usability of data collection tools, and systems interoperability were challenges cited in conducting maturity assessment in Australia, resonating with our findings.[32] In the US and Asia, context-based maturity assessments coupled with provider knowledge were key drivers to successful and sustainable assessments for their patient monitoring systems.[33,34] Our study's themes about routinising DHSMAs align with the US Public Health Informatics Institute's concept of using assessments for advancing US county-

different flower types, colors, and forms (Rubert, 1977) and are among the ornamental plants whose demand has increased day by day. Pot roses rank among the top two pot flowers and are worth approximately 14.3 million euros, and 15.7 million pieces of pot miniature roses were sold in worldwide pot flower sales according to Royal FloraHolland (2020) (Flora Cultural International, 2021). The increasing consumer demand for potted miniature roses motivates the breeders into the breeding programs and successfully improves many new miniature rose cultivars.

The development of new rose varieties involves high investment and labor. Therefore, there is a constant need for genetically improved breeding to combat conditions and for scientifically rigorous evaluation of inheritance processes to adequately document rose breeding. Crossing between closely related varieties with common ancestry obtains better results, which is commercially desirable; however, compatibility barriers may limit crossing success (Younis et al., 2014). Other problems include sterility and non-fertility of pollen donor parents. Therefore, breeders desire to integrate fertile parents into gene pools. Flower color, flower size, the number of petals, plant habit, fragrance, the number of flowers per stem, and flower longevity are the primary selection criteria in potted miniature rose breeding. Many rose cultivars have been developed with classical breeding approaches until today. These cultivars certainly ensure the criteria of the breeder for desirable traits, but gene crossing was limited. As a result, less variation in cytology-occurred compatibility decreased among candidate cultivars. Moreover, genetic barriers to hybridization are often an obstacle to obtaining interspecific hybrids in rose (Abdolmohammadi et al., 2014; Dogan et al., 2022). Furthermore, potted miniature roses have not a rich source of pollen for pollinators, and miniatures are known for their difficult sexual reproduction from pollination to seed set and germination (Datta, 2018). Therefore, although some researchers have reported that miniature roses got mini roses when crossed with climber, floribunda, cut flowers, or bush roses (Datta, 2018), in miniature rose breeding, less attention has been paid to scientifically rigorous studies on the quantitative and qualitative traits in their hybrids.

The present study aimed to cross the potential of some miniature rose varieties to compare crossability among the cultivars and the morphology of hybrid progeny.

2 Materials and methods

2.1 Material

The study was carried out on 25 May 2019 to 30 August 2022 in the rose breeding greenhouse, Department of Horticulture, Faculty of Agriculture, Ankara University (Turkey) (39°57′40.2″N 32°51′51.7″E). In this study, six different tetraploid commercial pot miniature roses (Juanita Kordana, Rosa Shining Star, Red Romance, Orange Romance, Rosa White Star, and Rosa Bling Love Star) were used as female parents, and a *R. hybrida* tetraploid cut rose cultivar (Black Baccara) and *Rosa centifolia* (tetraploid) (Dogan et al., 2020) were used as male parents (Table 1).

Twenty plants of each genotype were cultivated in single pots with perlite and cocopeat at the Ankara University. During the vegetation period, the temperature inside the greenhouse was kept 23 ± 5°C and the relative humidity between 60-70%. By using a 55% heat-shaded screen, the plants are prevented from being damaged by the intense light in summer. Water and fertilization were provided to the plants by drip irrigation system using a fertigation computer.

2.2 Method

2.2.1 Hybridization

Prior to hybridization, pollen parents were tested for pollen viability and germination *in vitro*. Anthers of cultivars were collected from the tight floral buds of 10 plants a day before. The pollen was taken into a glass bottle and kept at 24°C and 60%–65% humidity for 24 h in an incubator for dusting. The pollen viability was examined using the IKI (iodine potassium iodide) test (0.5 g of iodine and 1 g of potassium iodide; Eti, 1991). Pollen grains was spread on a drop of the solution on a coverslip. Four replicates and randomly chosen five fields were counted with a light microscope (×100). *In vitro* pollen germination of the cultivars was assessed using a germination medium (10 ppm (parts per million, (mg/L)) boric acid and 20% sucrose in 1% agar: Eti, 1990) using three petri dishes for each replicate. Four replicates were performed for each two pollen parents. Pollen was placed on the medium with a paintbrush, and, after 24 h of incubation at 24°C and 60%–65% humidity, pollen germination was evaluated. Pollen was assumed to germinate when a pollen tube reached a length of at least 1.5 times the pollen diameter (Leus, 2005; Dogan et al., 2020) with a light microscope (×100), and 250 pollen grains were calculated for each field.

Both pollen and seed parents were prepared for hybridization. Pollen grains were collected and stored in an incubator (22°C and 60%–65% humidity) for dusting, and seed parents were done emasculating when flower buds were in 50%–60% open stage in the morning to avoid self-pollination. Pollen was sprinkled on the stigma of emasculated plants with a brush in the morning, and the flower buds were covered in paper bags. Hybridization was made

TABLE 1 Nuclear DNA contents and ploidy levels of the roses in the study (Dogan et al., 2020; Dogan, 2022; Dogan et al., 2023).

Species/Variety	DNA (pg/2C)	Ploidy Level
Rosa Shining Star	2.37	Tetraploid
Red Romance	1.95	Tetraploid
Orange Romance	2.43	Tetraploid
Rosa White Star	2.39	Tetraploid
Rosa Bling Love Star	2.22	Tetraploid
Juanita Kordana	1.97	Tetraploid
Rosa centifolia	2.43	Tetraploid
Black Baccara	2.34	Tetraploid

of flowers per stem of progeny (all flowers were counted from the first bloom until the period just prior to pruning) was counted.

2.2.3 Statistical analysis

A randomized plot design with four replications was used to establish the experiment to ascertain pollen viability and germination rates. In each replication, an average of 250 pollen grains were counted in five randomly selected fields. Studies on crossbreeding were conducted using a randomized plot design with four replications. Five blooms were replicated in each cross, resulting in a total of 20 crosses for each combination. Data from pollen quality and crossbreeding studies were analyzed using IBM SPSS Statistics 26 software. Pairwise comparisons between all means were subjected to the Duncan's multiple range test, and differences were determined using analysis of variance. To define vigor-related effects on the parameters under study, better-parent (BP) and mid-parent (MP) heterosis estimations for the performance of F_1 progenies were made by the following formula (Fonseca and Patterson, 1968; Nadeem et al., 2015):

$$Ht\,\% = [(F_1 - BP)/BP] \times 100$$

$$Hbt\,\% = [(F_1 - MP)/MP] \times 100$$

3 Results

3.1 Pollen viability and germination, fruit sets, number of seeds, germination of seed

The Black Baccara and *R. centifolia* pollen were obtained as a bulk of at least 20 flowers (each one collected from a different plant of the same genotype) by gathering the anthers. *R. centifolia* showed the highest pollen viability percentage at 61.27% although the pollen germination rate was 32.19%. The percentage of the pollen viability of Black Baccara was 48.61%, whereas the percentage of pollen germination was 23.26%.

All the values of parameters such as fruit set, weight of fruit, total seed, weight of seed, seed per fruit, and germination of seed showed significant results within all crosses. The maximum percentage of fruit set was observed in crosses of Rosa White Star (65.0%–75.0%) (Table 2). A minimum success percentage of the fruit set (20.0%) was observed in the crosses of Juanita Kordana × Black Baccara, Orange Romance × Black Baccara, and Rosa Shining Star × *R. centifolia*. Orange Romance × *R. centifolia* as seed parents showed the highest value of weight of fruit (3.85 g) and weight of seed (0.74 g). The maximum total seed was obtained from the combinations in that Rosa White Star was the seed parent. The maximum number of seeds per fruit was produced by crosses of Rosa Bling Love Star × *R. centifolia* (12.63 seeds) followed by Red Romance × Black Baccara (12.50 seeds) and Orange Romance × *R. centifolia* (11.20 seeds). The maximum success germination of seed percentage was determined by Red Romance × Black Baccara (48.00%) and Orange Romance × *R. centifolia* (31.25%), respectively (Table 2).

All the parameters were correlated to forecast the degree of relationship with each other. The maximum value (r = 0.944**) for the positive correlation was recorded between the total seed and the weight of the hip. Although a strong correlation was examined between the total seed and the weight of fruit (r = 0.926**), between the weight of fruit and the weight of seed (r = 0.878**), and between the seed per fruit and the weight of fruit (r = 0.827**), respectively, the weakest correlation (r = 0.099) was examined between the fruit set and germination of seed. The percentage of fruit set had a strong correlation with the total seed (r = 0.654**), the weight of fruit (r = 0.481**), and the weight of seed (r = 0.481**) (Table 3).

TABLE 2 Comparison of different parameters in crosses of *Rosa* species.

Combination ♀ × ♂	FS (%)	WOF (g)	TS	WOS (g)	SPF	GOS (%)
Juanita Kordana × *R. centifolia*	30.0 cd	1.02 d	47.00 de	0.14 de	7.83 b	8.51 de
Juanita Kordana × Black Baccara	20.0 d	0.73 e	19.00 f	0.09 e	4.75 d	4.78 e
Red Romance × *R. centifolia*	35.0 cd	1.51 c	40.00 e	0.15 de	5.71 cd	6.25 de
Red Romance × Black Baccara	40.0 cd	3.20 a	100.00 b	0.49 a	12.50 a	48.00 a
Orange Romance × *R. centifolia*	50.0 bc	3.85 a	112.00 a	0.74 a	11.20 a	31.25 b
Orange Romance × Black Baccara	20.0 d	0.80 e	9.00 g	0.10 de	2.25 e	11.11 d
Rosa White Star × *R. centifolia*	75.0 a	3.10 a	132.00 a	0.68 a	8.80 ab	3.61 e
Rosa White Star × Black Baccara	65.0 ab	3.25 a	128.00 a	0.67 a	9.85 a	7.81 de
Rosa Bling Love Star × *R. centifolia*	40.0 cd	3.04 a	101.00 b	0.36 b	12.63 a	39.60 a
Rosa Bling Love Star × Black Baccara	45.0 c	2.78 a	91.00 c	0.24 c	10.11 a	32.97 b
Rosa Shining Star × *R. centifolia*	20.0 d	0.99 d	20.00 f	0.15 de	5.00 d	5.00 e
Rosa Shining Star × Black Baccara	35.0 cd	1.37 c	52.00 d	0.21 cd	7.43 bc	23.08 c

FS, fruit set; WOF, weight of fruit; TS, total seed; WOS, weight of seed; SPF, seed per fruit; GOS, germination of seed ($P \leq 0.05$). The letters indicate the differences between the means ($P \leq 0,05$).

TABLE 3 Correlation between some different parameters for cross-breeding in *Rosa* species.

Parameters	SPF	FS	WOF	WOS	TS	GOS
SPF	1.000					
FS	0.513**	1.000				
WOF	0.827**	0.718**	1.000			
WOS	0.650**	0.820**	0.878**	1.000		
TS	0.782**	0.797**	0.944**	0.926**	1.000	
GOS	0.757**	0.099	0.567**	0.227	0.402*	1.000

*, Significant (P ≤ 0.05); **, highly significant (P ≤ 0.01). FS, fruit set; WOF, weight of fruit; TS, total seed; WOS, weight of seed; SPF, seed per fruit; GOS, germination of seed.

3.2 Plant characteristics

There were significant differences in the height the of plant for the selected parents and progenies. The maximum heights (98 cm and 190 cm) were recorded for Black Baccara and *R. centifolia*, respectively. The excellent heights among F_1 progenies were determined for Rosa White Star × *R. centifolia* (25.40 cm), followed by Red Romance × Black Baccara (heights of 24.62 cm) (Table 4). All potted miniature rose parents have a maximum point for bush shape, although the maximum point of bush shape in F_1 genotype was that of Orange Romance × *R. centifolia* hybrids (1.57 quality point) (Tables 3, 4). Among the parents, Orange Romance, Rosa White Star, and Rosa Bling Love Star, which are among the potted miniature rose varieties, were observed with the highest score (quality points of 3) in leaf color, whereas Rosa White Star × Black Baccara hybrid was observed among the F_1 genotypes.

3.3 Flower characteristic

Significant differences (p ≤ 0.05) were exhibited by the parent genotypes and F_1 hybrid progenies for flower diameters. Whereas the largest flower diameter was recorded in *R. centifolia* (12.5 cm) from the parents, the largest average flower diameter in F1 genotypes was determined in the Rosa White Star × *R. centifolia* hybrid (6.92 cm). The number of petals varied between 25.0 (Orange Romance) and 95.0 (*R. centifolia*) among the parents; meanwhile, petals varied between 23.33 (Juanita Kordana × Black Baccara) and 39.39 (Rosa Bling Love Star × Black Baccara) among the F_1 genotypes. The presence or absence of fragrance was changed between quality points of 1 (all miniatures except for Orange Romance) and 3 (*R. centifolia*) among the parents when fragrance quality points changed from 1.0 to 2.0 points among the F1 genotypes. Among the F_1 genotypes, the number of flowers per plant ranged from 1.50 (Juanita Kordana × *R. centifolia*) to 2.80 (Rosa White Star × *R. centifolia*) (Tables 4, 5). The progeny of Rosa Shining Star × Black Baccara (12.25) exhibited the maximum flower persistence life, followed progeny of Red Romance × Black Baccara (12.20) (Table 4).

Exploitation in the F_1 progenies of the petal color was categorized as pink, red, orange, white, and yellow. F_1 progenies of different colored parents had 61.69% red, 31.84% pink, 3.98% orange, 1.49% white, and 1.00% yellow petals. Juanita Kordana × Black Baccara and Red Romance × Black Baccara combinations, in which both seed parent and pollen parents had red colored petals (70.83%–100%), were highly dominant in the red color among F_1 plants. Among the F_1 plants obtained from the combinations of parents with red and pink color petals (Juanita Kordana × *R. centifolia*, Red Romance × *R. centifolia*, Rosa Bling Love Star × Black Baccara, and Rosa Shining Star × Black Baccara), it was determined that the pink color was dominant to the red color. In F_1 progenies from the combination in which at least one of the parents was red (Orange Romance × Black Baccara, Rosa White Star × Black Baccara, Rosa Shining Star × Black Baccara), the color red was

TABLE 4 Morphological traits and flower characteristics of parents.

Cultivars/Species	NOPS	NOA	FD (cm)	HOP (cm)	NOPF	FRAG	BS	LC	FC
Juanita Kordana (P1)	54.0 cd	67.0 cd	4.5 d	27.5 b	67.0 b	1.0 a	3.0 a	2.0 a	Red
Red Romance (P2)	58.0 c	72.0 d	6.9 ab	25.5 b	36.0 d	1.0 a	3.0 a	2.0 a	Red
Orange Romance (P3)	80.0 b	139.0 c	7.2 a	26.5 b	25.0 e	2.0 a	3.0 a	3.0 a	Orange
Rosa White Star (P4)	65.0 c	65.0 d	6.6 b	25.1 b	32.0 d	1.0 a	3.0 a	3.0 a	White
Rosa Bling Love Star (P5)	46.7 d	218.0 a	6.0 c	27.5 b	45.0 c	1.0 a	3.0 a	3.0 a	Pink
Rosa Shining Star (P6)	62.7 cd	106.0 c	6.5 b	26.7 b	32.0 d	1.0 a	3.0 a	2.0 a	Yellow
Black Baccara (P7)	46.0 d	98.0 b	10.1 a	127.7 a	45.0 c	2.0 a	2.0 a	3.0 a	Dark Red
Rosa centifolia (P8)	160.0 a	190.0 ab	12.5 a	138.5 a	95.0 a	3.0 a	2.0 a	2.0 a	Pink

NOPS, number of pistil per flower; NOA, number of anthers per flower; FD, flower diameter; HOP, height of plant; NOPF, number of petal per flower; FRAG, fragrance; LC, leaf color; BS, bush shape; FC, flower color (P ≤ 0.05). The letters indicate the differences between the means (P≤ 0,05).

TABLE 5 Morphological traits and flower characteristics of F1 progenies.

Combination	FD (cm)	HOP (cm)	NOPF	FRAG	BS	LC	NOF	FPL
P1 × P8 (O1)	3.46 c	20.00 d	35.38 a	1.00 b	1.25 a	2.00 a	1.50 a	8.25 b
P1 × P7 (O2)	6.66 a	23.80 a	23.33 d	2.00 a	1.00 a	2.00 a	2.00 a	10.00 b
P2 × P8 (O3)	4.01 c	25.50 a	32.11 a	1.33 ab	1.33 a	2.33 a	2.45 a	9.50 b
P2 × P7 (O4)	5.04 bc	26.62 a	33.77 a	1.00 b	1.21 a	2.19 a	2.27 a	12.20 a
P3 × P8 (O5)	4.82 bc	24.82 a	34.21 a	1.27 ab	1.57 a	2.41 a	1.95 a	9.15 b
P3 × P7 (O6)	3.90 c	25.80 a	27.00 c	1.00 b	1.00 a	2.00 a	2.00 a	9.25 b
P4 × P8 (O7)	6.92 a	25.40 a	28.43 bc	1.60 ab	1.00 a	2.00 a	2.80 a	9.33 b
P4 × P7 (O8)	5.08 bc	23.15 ab	29.00 b	1.00 b	1.40 a	2.70 a	1.70 a	9.45 b
P5 × P8 (O9)	5.71 ab	21.30 cd	33.79 a	1.23 ab	1.25 a	2.63 a	2.73 a	9.15 b
P5 × P7 (O10)	4.12 bc	20.93 cd	39.39 a	1.00 b	1.15 a	2.45 a	2.52 a	10.15 b
P6 × P8 (O11)	4.00 c	21.95 bc	35.42 a	1.00 b	1.00 a	2.00 a	2.00 a	9.00 b
P6 × P7 (O12)	4.30 bc	24.15 a	32.15 a	1.00 b	1.33 a	2.56 a	2.25 a	12.25 a

FD, flower diameter; HOP, height of plant; NOPF, number of petal per flower; FRAG, fragrance; BS, bush shape; LC, leaf color; NOF, number of flower; FPL, flower persistence life ($P \leq 0.05$). The letters indicate the differences between the means ($P \leq 0,05$).

dominant to white, orange, and yellow colors. In F_1 plants from the combination in which at least one of the parents was pink (Orange Romance × *R. centifolia*, Rosa White Star × *R. centifolia*, and Rosa Shining Star × *R. centifolia*), the pink color was dominant to other color petals (Table 6).

Among the parameters, heterosis (Ht) and heterobeltiosis (Hbt) values varied. The heterosis was evident in 33.33% of the hybrid progeny of Juanita Kordana × Black Baccara (O2), for fragrance. The heterosis and heterobeltiosis were evident in 16.50% of the hybrid progeny of Red Romance × *R. centifolia* (O3), for leaf color (Table 7).

3.4 Cluster analysis

The data obtained as a result of the morphological characterization were evaluated by cluster analysis into three main clusters to reveal the similarity relationships of the

TABLE 6 Exploitation in the F_1 progenies of the petal color.

Progenies	Red (%)	Pink (%)	Orange (%)	White (%)	Yellow (%)
Juanita Kordana × *R. centifolia* (O1)	25.00	75.00	–	–	–
Juanita Kordana × Black Baccara (O2)	100.00	–	–	–	–
Red Romance × *R. centifolia* (O3)	33.33	66.67	–	–	–
Red Romance × Black Baccara (O4)	70.83	29.17	–	–	–
Orange Romance × *R. centifolia* (O5)	–	78.38	21.62	–	–
Orange Romance × Black Baccara (O6)	100.00	–	–	–	–
Rosa White Star × *R. centifolia* (O7)	–	100.00	–	–	–
Rosa White Star × Black Baccara (O8)	40.00	30.00	–	30.00	–
Rosa Bling Love Star × *R. centifolia* (O9)	–	100.00	–	–	–
Rosa Bling Love Star × Black Baccara (O10)	36.36	63.64	–	–	–
Rosa Shining Star × *R. centifolia* (O11)	–	100.00	–	–	–
Rosa Shining Star × Black Baccara (O12)	55.56	33.33	–	–	11.11
***Total**	**61.69**	**31.84**	**3.98**	**1.49**	**1.00**

* Color average of all F_1 plants.

TABLE 7 Heterosis and heterobeltiosis of F_1 hybrids for some parameters.

Progenies		FD	HOP	NOPF	FRAG	BS	LC
P1 × P8 (O1)	Ht (%)	−59.29	−78.31	−56.32	−50.00	−50.00	0.00
	Hbt (%)	−72.32	−87.00	−62.76	−50.00	−58.33	0.00
P1 × P7 (O2)	Ht (%)	−8.77	−71.54	−42.40	33.33	−60.00	−20.00
	Hbt (%)	−34.06	−82.93	−65.18	−33.33	−66.67	−33.33
P2 × P8 (O3)	Ht (%)	−58.66	−71.34	−50.98	−33.50	−46.80	16.50
	Hbt (%)	−67.92	−83.03	−66.20	−33.50	−55.67	16.50
P2 × P7 (O4)	Ht (%)	−40.71	−67.86	−16.62	−33.33	−51.60	−12.40
	Hbt (%)	−50.10	−80.72	−24.96	−50.00	−59.67	−27.00
P3 × P8 (O5)	Ht (%)	−51.07	−72.34	−42.98	−49.20	−37.20	−3.60
	Hbt (%)	−61.44	−83.52	−63.99	−57.67	−47.67	−19.67
P3 × P7 (O6)	Ht (%)	−54.91	−69.13	−22.86	−50.00	−60.00	−33.33
	Hbt (%)	−61.39	−81.36	−40.00	−50.00	−66.67	−33.33
P4 × P8 (O7)	Ht (%)	−27.54	−68.95	−55.23	−20.00	−60.00	−20.00
	Hbt (%)	−44.64	−81.66	−70.07	−46.67	−50.00	−33.33
P4 × P7 (O8)	Ht (%)	−39.16	−72.32	−24.68	−33.33	−44.00	−10.00
	Hbt (%)	−49.70	−83.44	−35.56	−50.00	−30.00	−10.00
P5 × P8 (O9)	Ht (%)	−38.27	−76.75	−51.73	−38.50	−50.00	5.20
	Hbt (%)	−54.32	−86.06	−64.43	−59.00	−58.33	−12.33
P5 × P7 (O10)	Ht (%)	−48.82	−75.61	−12.47	−33.33	−54.00	−18.33
	Hbt (%)	−59.21	−85.18	−12.47	−50.00	−61.67	−18.33
P6 × P8 (O11)	Ht (%)	−57.89	−75.85	−44.22	−50.00	−60.00	0.00
	Hbt (%)	−68.00	−85.60	−62.72	−66.67	−66.67	−33.33
P6 × P7 (O12)	Ht (%)	−48.19	−71.31	−16.49	−33.33	−46.80	2.40
	Hbt (%)	−57.43	−82.65	−28.56	−50.00	−55.67	−14.67

FD, flower diameter; HOP, height of plant; NOPF, number of petal per flower; FRAG, fragrance; BS, bush shape; LC, leaf color; Ht, heterosis; Hbt, heterobeltiosis.

genotypes. Most parents were grouped together with their progenies in a clustering method based on phenotypic (quantitative) characteristics (Figure 2). In general, miniature pot roses and progenies were in the same group.

4 Discussion

In breeding programs, the development of new rose varieties follows a sequence of processes such as fertilization, fruit set, seed formation, germination of seeds, and hybrid progeny after controlled pollination (Nadeem et al., 2013; Dogan et al., 2020). One of the most important steps in crossbreeding is the inclusion of fertile species and varieties in breeding programs. It is very important that the pollinator pollen, which is used as a male parent, is highly viable and germinating. In the studies carried out, the viability rates of some modern roses were found between 17.71% and 52.14% by Dogan (2022), and Pipino et al. (2011) reported that it ranged from 0% to 46.5%. The pollen germination rates of different hybrid tea roses were determined between 0% and 46.5% by Pipino et al. (2011), and Dogan et al. (2020) found that it varies between 24.61% and 45.24%. These studies show similarities with the present study. Although the fruit set, the number of seeds per fruit, and the seed germination rate of miniature roses are generally low, the miniature used as pollen parents and crossed with climbers, floribundas, and shrub roses is recommended (Datta, 2018; Dogan, 2022). In the present study, fertility of the parent cultivars ranged from low to high, most likely caused of multiple genetic factors, mitotic abnormalities, including lethal genes and their deleterious effects, incompatibility of gametes, the heterozygous polyploid, and interspecific factors of parent cultivars (Ogilvie et al., 1991). Overall, there was a satisfactory success rate of 39.58% among all possible crosses, which coincides with the findings of Dogan (2022) and Uran (2022) who observed that 30.3% and 37.92%, respectively, of all miniature crosses develop into mature hips, and the resulting seeds

Other covariates

Information was also collected on gender (male, female), age (54–91), years of education, and APOE gene (0, 1, 2 alleles).

Analytical strategy

We use used series of multinomial regression models predicting diagnosed cognitive status to assess whether additional covariates improved the variation explained in the model (as indicated by an increase in pseudo-R^2). In Model 1, we included covariates for sex, age, education and APOE4 genes (the control model). In Model 2, we add MMSE scores (MMSE only model). In Model 3, we substitute volumetric MRI markers for MMSE (MRI only model). In Model 4, we substitute CSF markers for MRI markers (CSF only model). In Model 5, we include all covariates (MMSE, MRI, and CSF) in a fully saturated model. These models allow us to examine the associations between each set of covariates and cognitive status. From model information, we are able to ascertain whether or not each set of covariates partly explains cognitive status and the potential contribution. For example, by comparing the pseudo-R^2 across models, we are able to determine whether certain sets of covariates explain more of the variation in cognitive status than others, and in the fully saturated model, we are able to evaluate the potential improvement in cognitive status classification when all sources of information are included (in our case, we sought to evaluate the improvement in cognitive status classification when MRI and CSF markers were included with MMSE scores: Model 2 compared to Model 5). In addition to these models, ancillary analysis evaluated each covariate separately (a model for each MRI and CSF marker). We found similar patterns to the models with the clusters for each set of markers; therefore, for parsimony, we do not present these findings but instead focus on how the sets of covariates improve cognitive status classification above and beyond only considering cognitive performance examinations.

All models were estimated using STATA 15.1.

Results

The sample descriptive information is reported in Table 1. The respondents were 73.1 years of age on average. Men constituted 56% of the sample. On average, respondents had 16 years of education. A slight majority had one or two APOE4 alleles: 39.3% had 1 allele and 12.1% had 2 alleles. The average MMSE score was 27.2. And most of the respondents had some form of

Table 1. Sample characteristics (ADNI, N = 832).

	Mean	(s.d)	%
Age	73.1	(7.21)	
Male			56%
Education (Years)	16	(2.72)	
APOE4			
0			48.6%
1			39.3%
2			12.1%
MMSE	27.2	(2.67)	
Cognitive Status			
Cognitive Normal			19.0%
Cognitive Impairment without Dementia			60.6%
Dementia			20.4%

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Table 2. Coefficients for multinomial logistic regression model predicting cognitive status from MMSE, MRI, and CSF markers (ADNI, N = 832).

	Model 1	Model 2	Model 3	Model 4	Model 5
Dementia					
Age	0.0103	-0.0533*	-0.105***	-0.0241	-0.123***
Male	0.328	-0.348	0.434	0.386	-0.272
Education (Years)	-0.103*	0.150*	-0.102*	-0.0653	0.140*
APOE4 Allelles (Reference: 0 Alleles)					
1 Allele	1.623***	1.278***	1.278***	0.684*	0.824*
2 Alleles	2.729***	1.477*	1.925***	1.017*	0.576
MMSE Score		-1.782***			-1.633***
Whole Brain Volume			0.340		0.411
Hippocampal Volume			-2.006***		-1.172***
Ventricle Volume			0.550***		0.373
Ptau				1.024***	0.693***
aß				-1.123***	-0.261
Constant	-0.160	48.68***	8.447***	2.057	50.17***
Cognitive Impairment Without Dementia					
Age	-0.0378**	-0.0636***	-0.0917***	0.0562***	-0.117***
Male	0.338	0.0724	0.180	0.372	-0.104
Education (Years)	-0.0242	0.0534	-0.0323	-0.00917	0.0406
APOE Allelles (Reference: 0 Alleles)					
1 Allele	0.967***	0.809***	0.777***	0.552*	0.462
2 Alleles	1.349**	0.879	0.815	0.663	0.192
MMSE Score		-0.621***			-0.529***
Whole Brain Volume			0.617***		0.580***
Hippocampal Volume			-1.254***		-1.014***
Ventricle Volume			0.141		0.150
Ptau				0.586***	0.456**
aß				-0.182	-0.0348
Constant	3.740**	22.34***	8.337***	5.218***	24.54***
Observations	832	832	832	832	832
Pseudo R-squared	0.064	0.401	0.182	0.142	0.445

* $p < .05$

** $p < .01$, **$p < .001$

Source: ADNI data

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cognitive impairment: 60.6% had cognitive impairment without dementia and 20.4% had dementia.

We estimated a several multinomial logistic regression models in Table 2 to evaluate the association between each set of covariates with cognitive status and compare the increase in variance explained (pseudo-R^2) across the models. In Model 1, the results for the demographic controls and APOE4 alleles are presented. We found that more years of education are associated with a decreased risk of having dementia, and more APOE4 alleles are associated with increased risk. We did not find statistically significant associations for age or gender for dementia. For CIND, we found a positive association with age and APOE4. We did not find statistically significant associations of gender or education for CIND. Overall, the demographic controls and APOE4 explain about 6.4% of the variance in cognitive status (pseudo-R^2 = .064).

Next, we added MMSE scores. Corresponding to expectations, higher MMSE scores are associated with a decreased likelihood of having dementia or CIND. The explanatory power of the model also improved markedly. The pseudo-R^2 increased from .064 in Model 1 to .401 in Model 2, a 34.6% increase in the explanation of variance with the addition of MMSE to the demographics and APOE4 only model.

In Model 3, we sought to evaluate the association of MRI volumetric measures with controls for demographics and APOE4 with cognitive status. As expected, we found that hippocampal volume is negatively associated with dementia or CIND status: one standard deviation increase in hippocampal volume is associated with an 87% decrease in the likelihood of having dementia and a 71.5% decrease in the likelihood of having CIND (OR .13 and .28, respectively). For dementia only, we found that ventricle volume is positively associated with dementia status. We, however, did not find a statistically significant association with whole brain volume. In contrast, for CIND, we found a positive association for whole brain volume and no statistically significant relationship with ventricle volume. Overall, these models showed a modest improvement in explanation of variance: the pseudo-R^2 improved from .064 in Model 1 (control only) to .182 in Model 3 (MRI + Controls).

To evaluate the associations of CSF markers with cognitive status, we examined a model with CSF markers and controls. These results are presented in Model 4. Overall, we find strong associations between CSF markers and dementia (less so for CIND). P-tau has a positive association with CIND and dementia: an increase in P-tau is associated with increased likelihood of having some form of cognitive impairment. For aß, we only find a negative association with dementia (no association was found for CIND): an increase in aß is associated with a decrease in the likelihood of having dementia. Compared to the model with only controls, the inclusion of CSF markers improved the variance explained (.064 in Model 1 to .142 in Model 4). However, compared to earlier models with MMSE and MRI model, the improvement was the smallest.

To evaluate the potential improvements in prediction when including MRI and CSF markers compared to MMSE alone, we evaluated a model with all covariates. Compared to the model with MMSE only (Model 2), including MRI and CSF markers marginally improved the variance explained by 10%: the pseudo-R^2 improved from .401 in Model 2 to .445 in Model 5. This provides evidence of marginal improvements in prediction with inclusion of MRI or CSF biomarkers.

Lastly, in Table 3 we evaluated the differences between diagnosed and predicted prevalence from Model 2 (MMSE only) and Model 5 (MMSE+MRI+CSF). These models performed relatively similarly. Both Models 2 and 5 predicted greater levels of CIND (13% for Model 2 and 9.9% for Model 5) and lower levels of dementia (-2.2% for both models) than were diagnosed. The difference between diagnosed and predicted was greatest with CIND and smallest in dementia for both models. Comparing the models to one another, we find small improvements in predicting the correct prevalence across cognitive statuses; the estimates in the fully saturated model were more closely aligned with the diagnosed prevalence, and were driven by

Table 3. Differences of observed and predicted prevalence of cognitive status in ADNI study (N = 832).

Cognitive Status	Diagnosed	Model 2: MMSE Only	Model 4: MMSE + MRI + CSF	Δ between Observed & Predicted from Model 2	Δ between Observed & Predicted from Model 4
Cognitively Normal	19.0%	8.2%	11.3%	-10.8%	-7.7%
Cognitive Impairment without Dementia	60.6%	73.6%	70.4%	13.0%	9.9%
Dementia	20.4%	18.3%	18.3%	-2.2%	-2.2%

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Statistical analysis

Demographic data is summarized using descriptive statistics and expressed as mean with SD for variables with a normal distribution, as median and interquartile ranges (IQRs) for those without normal distribution, or as a percentage for categorical variables. Data between groups were compared using either an independent *t* test, Mann–Whitney *U* test, or χ^2 test, as appropriate. A Wilcoxon signed-rank test was used to determine differences in mean values from the initial and the follow-up DXAs. Factors associated with low BMD were analyzed using univariate logistic regression. Multivariate logistic regression was performed for variables with $P < 0.1$ from univariate analysis and are expressed as an adjusted odds ratio (OR). These statistical analyses were performed using IBM Statistics for Windows (version 21). Power analysis was based on the BMD of the lumbar spine at baseline and at the last follow-up using Stata (version 14, StataCorp). For all statistical analyses, $P < 0.05$ in a two-tailed test was considered significant.

Results

Patient characteristics

We considered the demographic and clinical data from medical records of a cohort of 60 pediatric patients with SLE in this retrospective observational study. The majority of the patients (52, 87%) were girls, yielding a female-to-male ratio of 6.5:1. The mean age of all patients at diagnosis was 11 ± 2.5 years. The disease duration from the diagnosis to the first DXA ranged from 2 months to 58 months and to follow-up BMD measurement ranged from 11 months to 87 months, respectively. The most common renal pathology was proliferative nephritis (52%). All patients had been receiving steroid treatment at the time of the first DXA with a cumulative steroid dose ranging from 1.9 g to 42.6 g. None of our patients had a history of fracture. Demographic information including disease activity, medications, and laboratory results are summarized in **Table 1**.

BMD at the hip and lumbar spine

Of the 60 patients who had a DXA scan performed for the first time, 24 had a BMD *z* score ≤ −2, demonstrating a prevalence of low BMD of 40%. Only 51 patients had a follow-up DXA scan completed. At follow-up, 28 patients had a BMD *z* score ≤ −2, which indicates an increased prevalence of low BMD to 55%. Comparison between the first and the follow-up DXA showed a significant decrease in both hip BMD *z* score (median difference −0.25, 95% confidence interval [CI] −0.40 to −0.05; $P = 0.016$) and lumbar spine BMD *z* score (median difference: −0.49, 95% CI −0.69 to −0.28; $P < 0.001$) (**Figure 1**).

Table 1. Characteristics based on the BMD z score in pediatric patients with SLE

Characteristic	First DXA (n = 60)		*P*	Follow-up DXA (n = 51)		*P*
	BMD *z* score			BMD *z* score		
	> −2 (n = 36)	≤−2 (n = 24)		>−2 (n = 23)	≤−2 (n = 28)	
Age at time of DXA (years), mean (SD)	11.34 (2.42)	11.32 (2.72)	0.98†	10.62 (2.67)	11.38 (2.05)	0.28†
Disease duration (months), median (IQR)	13.5 (7–21)	17 (8.5–32.5)	0.39‡	26 (22–37)	40.5 (20.5–53.5)	0.31‡
Weight for age *z* score, mean (SD)	1.05 (1.64)	−0.12 (1.78)	0.014†*	0.92 (1.98)	0.16 (1.56)	0.15†
Height for age *z* score, mean (SD)	−0.58 (1.46)	−1.00 (1.01)	0.24†	−0.58 (1.07)	−0.92 (1.19)	0.32†
SLEDAI score at DXA, mean (SD)	5.21 (5.73)	6.29 (4.94)	0.48†	4.25 (2.90)	5.89 (4.24)	0.25†
C3 level, mean (SD)	89.31 (29.30)	88.11 (36.16)	0.89†	93.40 (21.10)	99.31 (28.10)	0.40†
Vitamin D level, median (IQR)	20 (8–29)	30.50 (13.5–60)	0.26‡	24.5 (19.6–33.1)	20.1 (13.3–29.2)	0.41‡
Cumulative corticosteroid dose (g), mean (SD)	11.70 (7.00)	17.38 (13.58)	0.10†	15.91 (6.19)	23.65 (14.62)	0.059†
Immunosuppressant, n (%)	25 (69.4)	17 (70.8)	0.53§	18 (78.3)	23 (82.1)	0.73§
Vitamin D supplement, n (%)	17 (47.2)	10 (41.7)	0.53§	13 (56.5)	20 (71.4)	0.61§
Calcium supplement, n (%)	19 (52.8)	7 (29.2)	0.16§	18 (78.3)	17 (60.7)	0.64§

BMD, bone mineral density; DXA, dual-energy X-ray absorptiometry; IQR, interquartile range; SD, standard deviation; SLE, systemic lupus erythematosus; SLEDAI, SLE disease activity index.

†Independent *t* test; ‡Mann–Whitney *U* test; §χ^2 test; *$P < 0.05$.

Factors associated with low BMD

There was no significant association between low BMD and age at the time of BMD measurement, disease duration, disease activity, vitamin D level, or the intake of either immunosuppressant drugs or vitamin D supplements. Mean weight for age *z* score in patients with low BMD was significantly lower than in patients with a BMD *z* score > −2 and was significant in univariate but not in multivariate analyses. The cumulative dose of steroids in patients with BMD score ≤ −2 tended to be higher than patients with a BMD *z* score > −2 (adjusted OR = 1.08, 95% CI 1.0 to 1.17; $P = 0.050$); however, this tendency was not significant **(Table 2)**.

Discussion

Children with SLE are susceptible to bone loss, osteoporosis, and fragility fracture due to disease development before reaching their peak bone mass. The high frequency of osteopenia in juvenile-onset SLE has been addressed in a few studies [7–9, 23]. Bone fractures are more likely to occur in SLE patients. A high frequency of vertebral fractures of up to 23% has been reported for patients with juvenile-onset SLE [24]. We defined low BMD for age as a *z* score −2 SD and found a high prevalence of low BMD in children with SLE of 40% with an increasing prevalence of 55% over follow-up. No patients whose data were included in the present study had a previous fragility fracture. The significant decrease in BMD *z* score at the lumbar spine found during follow-up was consistent with other studies, which indicated corticosteroids had an effect on the metabolism of bone, leading to bone loss, which occurs preferentially in trabecular bone, like lumbar vertebrae [8, 25, 26]. Some studies have found the cumulative corticosteroid dose was significantly associated with reduced BMD [7–9], while Castro et al. found no significant correlation [27]. The present study suggested a tendency for the cumulative dose of steroids to be higher in patients with low BMD; however, the tendency was not significant.

Optimal intake of calcium and vitamin D is crucial for the healthy mineralization of the skeleton. Vitamin D insufficiency and low levels of calcium contribute to reduced BMD and the appearance of fractures. Several factors lead to a greater risk of developing low levels of plasma vitamin D in patients with SLE, including low vitamin D intake, avoidance of sunshine, and the inflammatory process of the disease itself. However, we did not find a significant association between low BMD and vitamin D deficiency. There is a trend toward a lower risk of osteoporosis and fracture in patients who were treated with calcium and vitamin D, but the association was not significant. Despite unproven efficacy, calcium and vitamin D supplementation are recommended for all patients who undergo long-term use of steroids. The apparent lack of a significant association between low BMD and the following factors, such as disease activity, disease duration, and immunosuppressive drug use, may be attributed to the small sample size of patients whose data was included in the present study. This limitation corresponded to the low statistical power (calculated at 0.37)

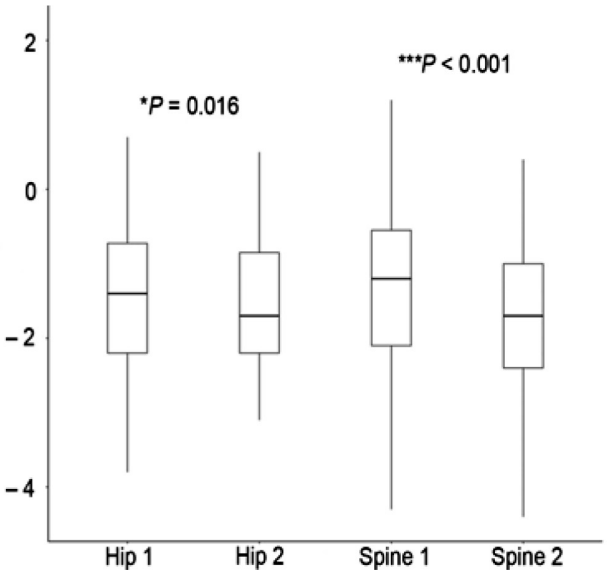


Figure 1. The difference in BMD *z* scores between the first and the follow-up DXA of the hip and lumbar spine. Hip 1, first DXA scan of the hip; Hip 2, follow-up DXA scan of the hip; LS 1, first DXA scan of the lumbar spine; LS 2, follow-up DXA scan of the lumbar spine. Wilcoxon signed-rank test; *$P < 0.05$, ***$P < 0.001$. BMD, bone mineral density; DXA, dual-energy X-ray absorptiometry.

Table 2. Factors associated with low BMD in patients with childhood SLE

Characteristic	First DXA				Follow-up DXA			
	Univariate analysis		Multivariate analysis		Univariate analysis		Multivariate analysis	
	OR (95% CI)	P†	Adjusted OR (95% CI)	P¶	OR (95% CI)	P†	Adjusted OR (95% CI)	P¶
Weight for age *z* score	0.67 (0.45–0.93)	0.03	0.78 (0.50–1.17)	0.25	0.87 (0.62–1.20)	0.394		
Cumulative steroid dose	1.06 (0.99–1.13)	0.10	1.00 (1.00–1.00)	0.10	1.07 (0.99–1.16)	0.059	1.08 (1.00–1.17)	0.050

Weight for age *z* score and cumulative steroid dose were included in the forward stepwise model of multiple logistic regression.
BMD, bone mineral density; CI, confidence interval; DXA, dual-energy X-ray absorptiometry, OR, odds ratio; SLE, systemic lupus erythematosus.
†Independent *t* test; ¶stepwise multiple logistic regression.

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immune checkpoint inhibitors-related myocarditis (ICIM) (n = 1). Baseline clinical characteristics and echocardiographic parameters are presented in Table 1.

Table 1
Characteristics of NICM patients.

	All patients (n = 29)
Clinical characteristics	
Male/Female	12/17
Age (years)	43.14 ± 16.94
cTnI (pg/ml)	128.63 ± 168.39
CK (U/L)	106.83 ± 125.79
CK-MB (μg/L)	2.77 ± 2.99
NT-proBNP (pg/ml)	1819.36 ± 2660.61
Disease duration (months)	41.00 ± 35.00
Baseline echocardiographic parameters	
LVEDD (mm)	60.79 ± 10.78
LVESD (mm)	47.69 ± 12.45
LVEF (%)	42.93 ± 14.66
^{68}Ga-FAPI-04 uptake	
SUV_{max}	4.16 ± 2.75
SUV_{mean}	2.09 ± 1.31
SUVR	1.92 ± 1.18
LVMV (mL)	196.13 ± 64.93

Values are mean ± SD for continuous variables. cTnI, cardiac troponin I; CK, creatine kinase; CK-MB, creatine kinase-MB; NT-proBNP, N-terminal fragment of pro-hormone brain natriuretic peptide; LA, left atrium; LVEDD, left ventricular end diastolic diameter; LVESD, left ventricular end systolic diameter; LVEF, left ventricular ejection fraction; SUV, standard uptake value; LVMV, left ventricular metabolism volume.

22 (75.9%) patients showed heterogenous elevated ^{68}Ga-FAPI-04 uptake in the left ventricle. Within them, 10 (34.5%) patients showed slightly diffuse activity in the right ventricle, including 1 RCM patient, 2 DCM patients, 3 HCM patients and 4 IC patients. The myocardial uptake in ^{68}GaFAPI04-avid patients (n = 22) was SUV_{max}=4.16 ± 2.75, SUV_{mean}=2.09 ± 1.31, SUVR = 1.92 ± 1.18, along with the diverse radioactive volume of LVMV = 196.13 ± 64.93, respectively. Since patient numbers were not sufficient to compare

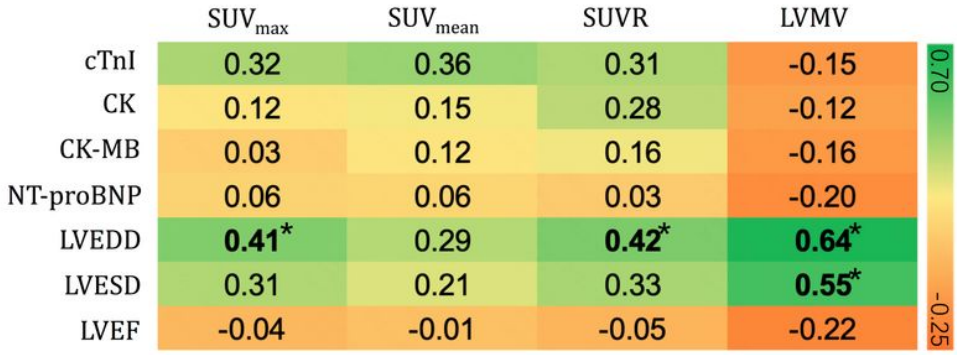


Figure 4

Correlation heatmap between ^{68}Ga-FAPI-04 uptake with clinical and echocardiographic parameters. *bold: P<0.05. cTnI, cardiac troponin I; CK, creatine kinase; CK-MB, creatine kinase-MB; NT-proBNP, N-terminal fragment of pro-hormone brain natriuretic peptide; LVEDD, left ventricular end diastolic diameter; LVESD, left ventricular end systolic diameter; LVEF, left ventricular ejection fraction; SUV, standard uptake value; LVMV, left ventricular metabolism volume.

TABLE 1 Included paper characteristics.

Authors	Title	Study design	Study period	Country	Primary outcomes	*N*	Vaccine type
Lee et al. (November 2022)[10]	COVID-19: Third dose booster vaccine effectiveness against breakthrough coronavirus infection, hospitalizations and death in patients with cancer: a population-based study	Population-based test-negative case-control design	8/12/2020–7/12/2021	UK	Breakthrough infections, hospitalization and death	Cancer: 2258553 Control: 84,781,190	Pfizer BioNTech AstraZeneca
Lee et al. (May 2022)[25]	Vaccine effectiveness against COVID-19 breakthrough infections in patients with cancer (UKCCEP): a population-based test-negative case-control study	Population-based test negative case control	8/12/2020–15/10/2021	UK	Breakthrough infections	Cancer: 377 194 Control: 28,010,955	Pfizer BioNTech AstraZeneca
Di Fusco et al.[26]	Evaluation of COVID-19 vaccine breakthrough infections among immunocompromised patients fully vaccinated with BNT162b2	Retrospective	10/12/2020–8/07/2021	US	Breakthrough infections, hospitalization and death	IC: 225,796 Control: 1,051,951	Pfizer BioNTech
Wang et al.[22]	Breakthrough SARS-CoV-2 Infections, Hospitalizations, and Mortality in Vaccinated Patients With Cancer in the US Between December 2020 and November 2021	Retrospective cohort study	12/2020–11/2021	US	Breakthrough infection, hospitalization and death	Cancer: 45,253 Control: 591,212	Moderna Pfizer-BioNTech Janssen/Johnson & Johnson
Liu et al.[27]	Risk Factors Associated With SARS-CoV-2 Breakthrough Infections in Fully mRNA-Vaccinated Individuals: Retrospective Analysis	Observational retrospective analysis	01/2021–09/2021	US	Breakthrough infection	IC: 7746 Control: 9139	Pfizer-BioNTech Moderna
Agrawal et al.[23]	Severe COVID-19 outcomes after full vaccination of primary schedule and initial boosters: pooled analysis of national prospective cohort studies of 30 million individuals in England, Northern Ireland, Scotland, and Wales	Prospective cohort	08/12/2020–28/02/2022	UK	Severe outcomes (hospitalization and death)	IC: 7754880 Control: 8,416,050	Pfizer-BioNTech AstraZeneca Moderna
Cox et al.[24]	Risk prediction of covid-19 related death and hospital admission in adults after covid-19 vaccination: national prospective cohort study	Prospective cohort	0/12/2020—15/06/2021	UK	Severe outcomes (hospitalization and death)	Total vaccinated cohort: 6,952,440	Pfizer-BioNTech AstraZeneca

TABLE 2 Calculated relative risk of breakthrough infections in the cancer cohort compared the non-cancer cohort.

The relative risk of breakthrough infections in the cancer cohort compared with the non-cancer cohort			
Authors	**Vaccine type**	***N***	**RR**
Lee et al. (November 2022)	Pfizer BioNTech AstraZeneca	2,258,553	1.73
Lee et al. (May 2022)	Pfizer BioNTech AstraZeneca	377,194	0.55
Di Fusco et al.	Pfizer BioNTech	81	1.84
Wang et al.[a]	Moderna Pfizer-BioNTech Janssen/Johnson & Johnson	45,253	1.24
Liu et al.	Pfizer-BioNTech Moderna	2400	1.62
Average			1.40

[a]Ratio of probability in patients with cancer versus non-cancer patients.

TABLE 3 Calculated relative risk of breakthrough infections in non-cancer immunocompromised cohort compared with non-immunocompromised cohort.

The relative risk of breakthrough infections in non-cancer immunocompromised cohort compared to non-immunocompromised cohort						
Authors	**Vaccine type**	**RR/N**	**Organ transplant**	**HIV**	**Chronic kidney disease**	**Immunodeficiency**
Di Fusco et al.	Pfizer BioNTech	RR	11.62	2.28	2.95	3.50
		N	687	2103	41,597	3190
Liu et al.	Pfizer-BioNTech Moderna	RR	2.25	1.56	1.56	1.60
		N	376	487	1514	2594
Average			6.94	1.92	2.26	2.55

Note: Patient definitions are available in the original papers, please refer for more information.

TABLE 4 Relative risk of hospitalization and death in cancer patients.

The relative risk of hospitalization and death in cancer patients				
Authors	**Vaccine type**	***N***	**Hospitalization**	**Death**
Lee et al. (November 2022)	Pfizer BioNTech AstraZeneca	2,258,553	0.92	3.21
Wang et al.[a]	Moderna Pfizer-BioNTech Janssen/ Johnson & Johnson	45,253	1.23	2.43
Average			1.08	2.82

[a]Used ratio of probability in patients with cancer versus non-cancer patients, immunocompromised group included only cancer patients. The ratio of probability was taken directly from the original paper due to a lack of data to calculate relative risk.

hospitalization, and death from SARS-CoV-2 compared with the healthy population.

3.3 | Chronic kidney disease

Several of the included papers reported on the susceptibility of chronic kidney disease patients to breakthrough infections and severe COVID-19-related outcomes. Generally, patients with chronic kidney disease had an average calculated risk of 2.26 (1.56–2.95) of COVID-19 breakthrough infection, despite vaccination (Table 3). The vulnerability of these patients was further demonstrated in the findings of Agrawal et al.[23] (Table 5) who reported adjusted risk ratios (aRR) of hospitalization and death of patients with chronic kidney disease. They reported aRRs of 1.51, 3.22, and 3.72 for stages 3, 4, and 5 of CKD, respectively (Table 5).

Supported by measures of both breakthrough infection and severe COVID-19 outcomes, it is evident that patients with chronic kidney disease remain vulnerable in the postvaccination era.

TABLE 5 Adjusted rate ratios of hospitalizations and death in immunocompromised patients taken from Agrawal et al.

Condition	*N*	aRR
Cancer		
Blood/bone marrow	114,300	3.35
Lung/oral	42,560	2.53
Transplants		
Solid organ	11,130	23.35
Bone marrow	4750	6.61
Renal	1120	9.87
Immunosuppressants	188,310	5.8
CKD		
Stage 3	171,860	1.51
Stage 4	6910	3.22
Stage 5	7180	3.72
Chemotherapy	76,060	2.7

Note: Please refer to the original paper for definitions of the following IC conditions. aRRs taken directly from the original paper. Patient definitions are available in the original paper, please refer for more information.

Abbreviation: aRR, adjusted risk ratio.

3.4 | Other conditions

We identified two studies which reported on breakthrough infection in fully vaccinated individuals with immunodeficiency and individuals with HIV as summarized in Table 3. Fully vaccinated individuals with immunodeficiency were at higher risk of breakthrough infection when compared to the healthy control population with an average relative risk of 2.55.

Similarly, patients with HIV were also at higher risk of breakthrough infection when compared to the healthy population with an average relative risk of 1.92. Not surprisingly, patients treated with immunosuppressants, and chemotherapy are at higher risk of severe COVID-19 outcomes with adjusted rate ratios for hospitalization and death for immunosuppressants and chemotherapy reported at 5.8 and 2.7, respectively.

4 | DISCUSSION

This systematic review assessed the existing data describing the efficacy of COVID-19 vaccination in protecting IC individuals against breakthrough infections and severe COVID-19 outcomes. These individuals are also more likely to experience severe complications from these breakthrough infections and they do not mount the same immune response following vaccination.[6–9] Whilst our results showed that IC patients are at increased risk of breakthrough infection, hospitalization and death (Tables 2–5), this risk differed between different sub-groups. Organ transplant patients were found to be the most vulnerable in this group with regard to breakthrough infection (RR = 6.94), then immunodeficient patients (RR = 2.55), followed by patients with chronic kidney disease (RR = 2.26), patients with HIV (RR = 1.92), and patients with cancer (average RR = 1.40) (Tables 2 and 3). Similarly, organ transplant patients were found to be the most vulnerable with regard to hospitalization and death postvaccination as this group demonstrated the highest adjusted rate ratios for hospitalization and death with solid organ transplants at highest risk (aRR = 23.35), followed by renal transplants (aRR = 9.87) and bone marrow transplants (aRR = 6.61) (Table 5). An increase in the risk of hospitalizations and death postvaccination can be observed with the increasing stages of CKD with stage 5 CKD (Table 5). The differences in relative risk between different sub-groups of IC patients highlights important clinical considerations in the management of these patients. This may involve careful consideration regarding vaccination boosting strategies and public health interventions. Different cancer subtypes were found to carry different risks for breakthrough infection and severe COVID-19 outcomes. Hematological cancer patients, particularly lymphoma, were consistently found to be a high-risk group amongst included papers,[10,22,25] with higher reported adjusted rate rations for severe COVID-19 outcomes for blood and bone marrow cancer when compared to lung and oral cancer (Table 5). Interestingly, Liu et al.[27] reported that cancer patients with a history of tumors did not exhibit a significant increase in risk suggesting that individuals whose cancers were in remission have a similar risk to the average population. With this, it seems that patients with active cancer and are undergoing active treatment are the most vulnerable.

Lee et al.[25] reported that cancer patients had a reduced risk of breakthrough infection when compared to the general population. Whilst breakthrough infection is a metric of vaccine effectiveness, it carries more limitations than metrics such as COVID-19-related hospitalizations and death. Breakthrough infection reflects behavior as well as vaccine effectiveness, and, a shielded population (individuals who are clinically vulnerable and at high risk of severe COVID-19 outcomes, for example, the IC cohort and are therefore isolating to reduce their risk of infection) may be less likely to encounter the virus. Therefore, this must be carefully considered when interpreting breakthrough infection as a metric for vaccine effectiveness. The waning immune response to the COVID-19 vaccine has

1 Introduction

During the transition period, dairy cows experience an imbalance in bioenergetics due to the higher energy demands for milk synthesis and maintenance, which exceed dietary intake, resulting in a negative energy balance (NEB) (1). This imbalance occurs when the dietary supply of gluconeogenic precursors fails to meet the liver's glucose output required by the mammary glands for milk production (2). Consequently, the body begins to catabolize amino acids from skeletal muscle and glycerol from adipose tissue to compensate for the energy deficit (3). Furthermore, reduced insulin activity due to NEB triggers lipolysis and the release of nonesterified fatty acids (NEFA) and beta-hydroxybutyric acid (BHBA) which are partially oxidized in the liver to produce ketone bodies (4). This process spares glucose for milk synthesis by inducing insulin resistance in muscle and adipose tissues (5). These metabolic adaptations, including glucose-sparing mechanisms, arise from the temporal mismatch of NEB and the compromised support for milk production. This energy deficiency increases susceptibility to metabolic disorders, including ketosis, fatty liver, and displaced abomasum in transition dairy cows (6). Additionally, research has shown that NEB can lead to a higher incidence of inflammatory responses, such as mastitis and endometritis, which may result in infertility (7, 8).

Glucose supplementation is critically important during the transition period to facilitate maximal milk synthesis and support immune function (2, 4). Glucose serves as a precursor for milk lactose, which plays a key role in the osmoregulation of milk (9). Hepatic glucose circulates in the blood, maintaining glucose homeostasis within a narrow range to meet the needs of peripheral tissues, including the mammary glands, muscles, adipose tissue, and the central nervous system (10, 11). It is estimated that approximately 72 grams of glucose are required to synthesis 1 kg milk (12, 13), and under homeostatic control, glucose supply does not limit milk synthesis (14). However, during specific circumstances such as the transition period, immune activation, and heat stress, glucose supply may become a limiting factor for milk yield in dairy cows (2). Moreover, it has been observed that fresh cows also experience inflammatory responses due to the ongoing uterine involution period (15). Therefore, providing a dietary glucose source with minimal rumen fermentation, such as rumen-protected glucose (RPG), represents a safer and more effective nutritional strategy to enhance intestinal glucose uptake, supporting higher milk production and postpartum immune function in dairy cows (16). Dietary supplementation of RPG may provide a surge of circulating glucose to support the milk production of high-producing animals. Studies revealed that RPG had a mixed impact on feed intake and milk production of dairy cows (17, 18). These variations in results might be due to varying precursors of glucose in products (mostly sucrose), dosage, and coating technology (freeze-drying or simple melting techniques). We hypothesized that dietary RPG supplementation could optimize lactating dairy cows' production performance and health. The objectives of this experiment were to evaluate the influence of RPG levels on feed intake, production performance, milk quality parameters and selected circulating blood biomarkers in early lactating Holstein Friesian.

2 Materials and methods

2.1 Experimental site, research design, and farming practices

The protocols and procedures of this experiment were approved by the Animal Use and Care Committee of the University of Veterinary and Animal Science, Lahore, before the research trial (DAS/536: 12–02–2020). The trial was conducted at a commercial dairy farm (Nawab Dairy Farm, Khan Garh, Muzaffargarh; 30°4′10″N 71°11′39″E) to evaluate the influence of dietary supplementation with rumen-protected glucose (RPG) on production performance, milk quality, and health in early lactating cows. Thirty-two Holstein Friesian cows in early lactation (6 ± 1 days in milk; milk yield averaging 30 ± 5.0 kg/day; live body weight 550 ± 50 kg; and body condition score 3.00 ± 0.25) were randomized into four groups (n = 8 cows/group) and assigned to one of four RPG levels in a completely randomized design. The groups were fed a basal diet (Control) or a basal diet supplemented with 150 g (S-150), 300 g (S-300), or 450 g (S-450) of RPG. Menoglu-plus (containing 49% edible dextrose coated with 51% palm oil) was used as the RPG source (Menon, Shanghai, China). The product's glucose content was determined using the manufacturer's information, which included 90% rumen bypass and 90% intestine release rate, and the dosage was adjusted to assure accurate glucose supplementation for each group. The maximum dose was included after considering for the incidence of insulin resistance. The RPG supplementation was top-dressed to the basal diet of the respective group with receptive dosage. The basal diet was a total mixed ration and its detailed formulation and chemical compositions are provided in

TABLE 1 Basal diet feed formulation on a dry basis.

List	Inclusion level
Corn silage (%)	32.8
Rhodes grass hay (%)	7.0
Alfalfa haylage (%)	2.5
Wheat straw (%)	1.89
Corn grain (%)	24.0
Soybean meal (%)	12.5
Canola Meal (%)	5.5
Rapeseed meal (%)	3.2
Corn gluten meal 60% (%)	2.5
Sugarcane molasses (%)	4.0
Calcium carbonate (%)	0.6
Di-calcium phosphate (%)	0.2
Sodium bicarbonate (%)	0.5
Sodium chloride (%)	0.2
Mineral Mixture[1] (%)	0.7
Urea (%)	0.208
Bypass fat-84 (%)	1.63
Toxin binder (clay base) (%)	0.052
Live Yeast (20 billion CFU/g) (%)	0.02

[1]Vitamin & Premix mix contained 9.50% Ca, 10.20% P, 1.43% Mg, 2.50% K, 2.40% S, 4.00% Na, 6.90% Cl, 0.60% Fe, 1.80% Zn, 0.15% Cu, 0.17% Mn, 0.005% Se, 0.002% Co, 0.02% I, 800 IU/g of vitamin A, 200 IU/g of vitamin D, and 12.40 IU/g of vitamin E.

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TABLE 2 Basal diet compositional characteristics on a dry basis.

Nutrient composition	
Variable	Value
Dry matter (%)	53.21
Crude protein (%)	17.48
Neutral detergent fiber (%)	28.41
Acid detergent fiber (%)	15.89
Non-fibrous carbohydrates (%)	45.24
Ether extract (%)	4.54
Predicted values[7]	
Metabolizable protein (g/kg of DM)	121.0
RUP[1] (% CP)	41.01
RDP[2] (% CP)	58.99
ME[3] (Mcal/kg of DM)	2.84
NEL[4] (Mcal/kg of DM)	1.79
DCAD[5] (mEq/kg)	209.0

[1]RUP, rumen undegradable protein.
[2]RDP, rumen degradable protein.
[3]ME, metabolizable energy.
[4]NEL, net energy for lactation.
[5]DCAD, dietary cation-anion difference.
[7]predicted values were calculated by NDS-RUMIN software.

Tables 1, 2 respectively. The analysis of dietary ingredients and the basal diet was performed for dry matter, crude protein (method 984.13, N × 6.25; Kjeldahl method), and ether extract (method 920.39) following the official procedures of Baur and Ensminger (19). Neutral detergent fiber (α-amylase + sodium sulfite-treated filtration) and acid detergent fiber (sulfuric acid + cetyltrimethylammonium bromide-treated filtration) were analyzed sequentially using the Ankom-2000 fiber analyzer (Ankom Technology Corp., Fairport, NY, United States). After a two-week dietary adaptation period, a 40-day feeding trial was conducted. All cows were housed in a free-stall system equipped with ventilation fans. Cows were fed twice daily (6:00 and 18:00 h) with the basal diet mixed as a total mixed ration in a wagon mixer (Dunker T1-80, Storti SPA, Italy). Sand bedding was provided in stalls, plowed daily in the morning, and replaced as needed. Manure was scraped twice daily (4:00 and 19:00 h) using pressure water and a manual disc. Cows were milked twice daily (7:00 and 19:00 h) in the milking parlor using an automatic milking system. After milking, udders were washed, cleaned, and disinfected, with teats dipped following standard operating procedures.

2.2 Body weight and feed intake

Cows' live body weights were recorded before the experiment and then weighted weekly using a digital weighing balance to estimate the changes in body weight. Feed was offered twice (6:00 and 18:00 h) a day and orts were recorded the next day before morning feeding to calculate the dry matter intake (DMI). The DMI was calculated using the following Equation 1.

$$DMI\ (kg) = \left(\left(\text{Feed offer}\ (\text{kg}) - \text{orts}\ (\text{kg})\ \text{X DM}\ of\ feed\right)\right) / number\ of\ cows\ in\ group \quad (1)$$

2.3 Milk volume and compositional characteristics

The milk volume of each cow was recorded daily using a digital weighing balance. Following a sterile protocol, approximately 30 and 20% of the total milk volume were sampled in the morning and evening, respectively. The collected samples were composited, and each cow's subsample was taken daily to create a representative sample. These samples were then analyzed for milk composition using a Lactoscan milk analyzer (Lac-toscan Farm Eco, Milkrotronic, Nova Zagora, Bulgaria). Energy-corrected milk, fat-protein-corrected milk, and the milk-to-feed ratio were calculated using Equations 2–4, respectively.

$$ECM\ (kg) = \begin{pmatrix} (0.325\ \text{X milk yield}\ (\text{kg})) + \\ (12.86\ \text{X milk fat yield}\ (\text{kg})) + \\ (7.04\ \text{X milk TP yield}\ (\text{kg})) \end{pmatrix} \quad (2)$$

$$FPCM\ (kg) = (0.4\ \text{X Milk Yield}\ (\text{kg})) + (10\ \text{X Fat Yield}\ (\text{kg})) + (7\ \text{X Protein Yield}\ (\text{kg})) \quad (3)$$

$$\text{Milk to Feed ratio} = (\text{ECM}\ (\text{kg}) / \text{DMI}\ (\text{kg})) \quad (4)$$

2.4 Blood biochemical analysis

Weekly blood samples of all cows were collected from their coccygeal vein 4 h after morning feeding by following a sterile method. Blood samples were collected in non-EDTA-containing vacutainers, and serum was separated by centrifugation at 2,000 × g for 15 min. The harvested serum samples were stored in labeled tubes at −20°C until further analysis. The serum was analyzed for BHBA, NEFA, and glucose, using commercially available enzymatic kits (Biosystems, Barcelona, Spain) by following the manufacturer's instructions.

2.5 Statistical analysis

The collected data were subjected to QQ plots and log transformation (SAS v. 9.4, University Edition, SAS In-stitute Inc., Cary, NC, United States) to test for normality and find the missing values. Data on production performance were analyzed using the MIXED procedure of SAS (SAS v. 9.4, University Edition, SAS Institute Inc., Cary, NC, United States). Individual cows were considered as the experimental units. The model included the following fixed effects: treatments, weeks, and the interaction of treatments × weeks. The statistical model was:

$$\text{Yijkl} = \mu + \text{Tj} + \text{Wk} + \text{TWjk} + \varepsilon\text{ijkl},$$

where, Yijkl = observation, μ = population mean, Tj = treatment effect, Wk = Weeks effect, TWjk = interaction of treatment and Weeks, εijkl = residual error. For multiple observations across weeks, the autoregressive [AR (1)] structure was applied within the framework of the repeated measures based on Akaike information criterion

values. Blood metabolites were analyzed using GraphPad Prism (version 9.0). Significance was declared at $p < 0.05$, and means were compared using Tukey's test. Weekly raw data values of milk yield were subjected to broken line regression using GraphPad Prism (version 9.0) for dosage-to-yield curve fit analysis with the following model.

$$Y = L + U \times (R - X) \times 1$$

Where: Y = dependent variable, L = theoretical maximum, R = requirement, X = independent variable, I = 1 (if X < R) or I = 0 (if X > R), U = rate constant.

3 Results

3.1 Production performance

According to Table 3, dietary supplementation with RPG significantly influenced the production performance parameters of early lactating cows. Final body weight increased linearly with RPG supplementation; with cows receiving 450 g/day RPG showing the greatest ($p < 0.05$) increase across the groups. Similarly, milk yield, energy-corrected milk, and fat-protein-corrected milk were significantly greater ($p < 0.05$) in the supplemented groups compared to cows fed the control diet. There was a significant treatment-by-week interaction for these milk production parameters, indicating that milk yield improved with increasing days in milk ($p < 0.05$). The milk-to-feed ratio of early lactating cows also improved significantly ($p < 0.05$) with RPG supplementation, while feed intake remained similar ($p > 0.05$) across the groups. The broken line analysis revealed that dietary dosage of RPG @ 450 improved weekly milk yield ($p < 0.05$), suggesting a dire need to explore the upper levels like 500 and 550 g/cow/day (Figure 1).

3.2 Milk compositional characteristics

The milk protein and milk fat contents of cows were not significantly influenced ($p > 0.05$) by dietary supplementation with RPG (Table 4). However, milk lactose, solids-not-fat, and total solids contents were significantly improved ($p < 0.05$) in the RPG-supplemented groups compared to the control group. There was also a significant treatment-by-week interaction between milk lactose and solids-not-fat contents. Milk lactose content was highest in cows supplemented with 450 g/day RPG, followed by 300 g/day RPG ($p < 0.05$). There was no significant difference in milk lactose content between the control and 150 g/day RPG-supplemented groups ($p > 0.05$). Solids-not-fat content increased linearly with increasing RPG supplementation ($p < 0.05$). Similarly, total solids contents were greater in the S-300 and S-450 supplemented groups compared to the other treatments ($p < 0.05$).

3.3 Blood biochemical analysis

The blood glucose, NEFA, and BHBA concentrations of dairy cows were significantly influenced ($p < 0.05$) by dietary supplementation with RPG (Figures 2–4). Blood glucose concentration increased linearly with RPG supplementation (Figure 2). The highest blood glucose concentration was observed in the S-450 group,

TABLE 3 Production performance of early lactating cows supplemented with different levels of rumen-protected glucose.

Variable	Treatments[1]				SEM[2]	*p*-value		
	Control	S-150	S-300	S-450		Trt[3]	week	Trt*week
FBW (kg/cow)	478.8^{d}	494.2^{bc}	501.4^{bc}	519.2^{a}	4.628	<0.0001	–	–
Feed intake (Kg/d)	27.07	29.28	28.81	28.93	3.246	0.125	0.131	0.151
Milk Yield (Kg/d)	23.88^{d}	27.05^{c}	28.23^{ab}	28.62^{ab}	0.196	<0.0001	<0.0001	<0.0001
ECM (kg/d)	21.48^{d}	22.93^{c}	23.16^{ab}	23.27^{ab}	0.137	<0.0001	<0.0001	0.010
FPCM (Kg/d)	21.07^{d}	22.68^{c}	22.96^{b}	23.16^{a}	0.120	<0.0001	<0.0001	<0.0001
Milk to feed ratio	0.76^{d}	0.81^{b}	0.80^{c}	0.82^{a}	0.005	<0.0001	<0.0001	<0.0001

[1]treatments = basal diet without supplementation (control), and basal diets with supplementation @ 150 g (S-150), 300 g (S-300), and 450 g (S-450) of rumen-protected glucose per day per cow.
[2]SEM, standard error of means.
[3]Trt, represents *p*-value for treatments.
The superscripts a–d without common sharing in a row mean a significant difference ($p < 0.05$), while common sharing among groups represents a non-significant difference ($p > 0.05$).

TABLE 4 Milk Composition of early lactating cows supplemented with different levels of rumen-protected glucose.

Variable	Treatments				SEM	*p*-value		
	Control	S-150	S-300	S-450		Trt	Week	Trt*week
Milk protein (%)	2.99	3.05	2.94	2.99	0.031	0.089	0.444	0.864
Milk fat (%)	4.10	4.25	4.22	4.21	0.052	0.207	0.025	0.906
Milk lactose (%)	4.33^{cd}	4.33^{cd}	4.36^{b}	4.37^{a}	0.009	0.011	0.245	0.002
Milk solid not fat (%)	7.87^{d}	7.90^{c}	7.94^{b}	7.95^{a}	0.016	0.003	0.506	0.045
Milk total solids (%)	11.98^{d}	12.15^{c}	12.17^{ab}	12.17^{ab}	0.055	0.044	0.022	0.634

The superscripts a-d without common sharing in a row mean a significant difference ($p < 0.05$), while common sharing among groups represents a non-significant difference ($p > 0.05$).

Table 1
The baseline information

	Control group (N = 217)	Experimental group (N = 219)	P value
Gender: male	86 (39.6%)	75 (34.2%)	0.244
Smoking status:yes	57 (26.3%)	53 (24.2%)	0.619
Pathological type: adenocarcinoma	200 (92.2%)	198 (90.4%)	0.516
Lung cancer stage: early	192(88.5%)	201(91.8%)	0.320
Age (years)	53 [48, 57]	52 [47, 57]	0.078
BMI (kg/m2)	23.5 [21.6, 25.7]	23.4 [21.4, 25.0]	0.285
D-dimer [a] (mg/L)	0.18 [0.12, 0.27]	0.17 [0.12, 0.26]	0.740
PT [a] (s)	10.3 [10.0, 10.7]	10.3 [9.9, 10.7]	0.331
APTT [a] (s)	26.8 [25.5, 28.1]	26.6 [25.4, 27.9]	0.250
FIB [a] (g/L)	2.62 [2.27, 3.01]	2.59 [2.33, 3.00]	0.792
TT [a] (s)	17.9 [17.4, 18.5]	18.0 [17.4, 18.5]	0.737
PLT [a] (10^9/L)	200 [167, 237]	202 [170, 246]	0.456
perimeter a (cm)	35.65 [34.10, 37.40]	35.58 [33.90, 37.21]	0.754
CTVR (%)	24.0 [20.0, 27.6]	24.0 [21.1, 27.7]	0.279
caprine b (points)	7 [7, 8]	7 [7, 7]	0.257

Values were presented as n (%) or median [IQR]. a indicated preoperative results; b indicated postoperative results.

Abbreviations: BMI: body mass index; PT: prothrombin time; APTT: activated partial thromboplastin time; FIB: fibrinogen; TT: thrombin time; PLT: platelet; CTVR: tube diameter/internal venous internal diameter; CVC: central venous catheter.

Primary Analysis

The median extubation time for CVCs in both groups was 6.00 days [5.00; 7.00] (P = 0.63). DVT was found by vascular ultrasound, specifically, 3 in the control group and 1 in the experimental group. CRT was produced in 173 (79.7%) control patients and 157 (71.7%) patients in the experimental group (p = 0.051). The multivariate Log-binomial regression showed that the risk of developing CRT in the experimental group was lower than in the control group (Adjusted RR = 0.889 [95% CI: 0.799–0.989], p =

0.031) (Table 2). One month after discharge, a total of 359 patients returned to the hospital for review, and 24 patients still had CRT, including 14 patients in the control group and 10 patients in the experimental group.

Table 2
Intervention RR was compared with the adjust RR

RR	95% CI	P value	Adjust RR	95% CI	P value
0.899	0.808–1.001	0.051	0.899	0.799–0.989	0.031

Exploratory Analyses

Multicollinearity tests suggested a possible collinearity between preoperative and postoperative Caprini scores and no collinearity among the remaining variables. Considering the actual situation, the postoperative Caprini variable is retained when the regression equation is constructed. The preoperative TT is adjusted in the log-binomial regression model as a confounding factor according to CIE method. According to the literature review, the four subgroups were set to D-dimer, FIB[21], CTVR[22], Caprini score[23], and age, smoking status, and extubation time were set to be the three exploratory subgroups for the interactive item subgroup analysis. The subgroup analysis showed that with D-dimer > 0.17mg/L, FIB < 2.59g/L, CTVR < 24%, and Caprini score > 7, the effect of CRT reduction by using IPC might be more significant, but the interaction term showed no significant difference in treatment effect between the groups (P for Interaction > 0.05). The findings were robust in the subgroup analyses (Fig. 2).

Furthermore, there were no significant differences in D-dimer, PT, APTT, TT, PLT, or leg circumference between the two groups at the follow-up, but in FIB was statistically significant (P = 0.019) (Fig. 3).

Discussion

This is the first prospective, randomized, controlled clinical trial in the Chinese population using LMWH + IPC antithrombotic therapy during the perioperative phase of lung cancer, especially for CRT caused by CVCs placed in lung cancer surgery, an easily neglected risk factor postoperation.

Although Chinese medical staff have a positive attitude toward the clinical application of mechanical prevention, the relevant knowledge and behavioralnorms need to be strengthened[24]. There are no relevant studies on whether IPC reduces CRT, and there are no specific guidelines or consensus for automated prevention, and the existing relevant policies and agreements are not detailed enough for mechanical prevention, leading to deficient feasibility. The guidelines recommended using IPC for more than 18h per day and an extension for completely inactive patients[25]. In consideration of patients' tolerance, machine availability, and the number of nurses, IPC was given twice a day for 30 minutes and maintained until the seventh day after surgery. Both groups were encouraged to move gently in bed or out of bed according to their postoperative recovery. We found that patients in the experimental group may

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enzymes, drug transporters, and drug receptors).[14] These genes are sometimes referred to as pharmacogenes. Other basic pharmacogenetic nomenclature with which healthcare providers should be familiar is shown in Table 1. Genetic variants can result in proteins with normal (i.e., population average), increased, decreased, or complete loss of function/expression. Specific examples of the clinical pharmacogenetic effects on cardiovascular drugs will be reviewed in more detail later on in this tutorial. However, in general, when patients have a genetic variant that results in decreased function of a drug metabolizing enzyme or transporter, then those patients can be exposed to higher drug concentrations systemically (at equivalent doses). When patients have a genetic variant that results in an increase in the function of a drug metabolizing enzyme or transporter, then those patients can be exposed to lower drug concentrations systemically (at equivalent doses). The clinical effects (i.e., increased toxicity or decreased efficacy) of these differences in systemic concentrations depends on whether the drug is activated or deactivated by metabolism. Some drugs, like clopidogrel, are prodrugs because they are activated by metabolism *in vivo*. Thus, the genetic variants would have the opposite clinical effects as typical drugs that are inactivated by metabolism. When patients have a genetic variant in the drug's target or receptor, then that could alter the receptor's sensitivity to the drug and therefore affect the drug's efficacy as well. Pharmacogenetic variants are quite common in the general population. Over 90% of patients carry at least one actionable pharmacogenetic variant, which is defined as a variant with clinically significant effects on risk of toxicity and/or therapeutic failure.[15–17]

The term "pharmacogenetics" was introduced around the end of the 1950s. A 1958 study showed significantly different blood concentrations of the antibiotic isoniazid in European and East Asian patients.[18] This was attributed to genetic variations in the drug's metabolism (variants in the N-acetyltransferase 2 gene). The term "pharmacogenomics" was coined more recently.[19] By the early 1990s, after decades of improvement in analytical methods and advances in our understanding of the human genome, the routine use of clinical pharmacogenetic testing was beginning to be supported.[20] During the past decade, clinical implementation of pharmacogenetics has significantly advanced, potentially in large part due to the development of clinical practice guidelines that provide evidence-based pharmacogenetic test result interpretations and recommendations for select gene-drug pairs.[21]

Today, pharmacogenetic evidence has reached almost every therapeutic area (e.g., oncology, cardiology, neurology, psychiatry, gastroenterology, and infectious diseases). Although some academic medical centers have spearheaded clinical implementation efforts, other health systems are gradually adopting the practice. Initial pharmacogenetic clinical implementation programs used reactive pharmacogenetic testing,[22] but the trend has begun to shift mostly to pre-emptive pharmacogenetics testing in the last decade because of increased recognition of the advantages and feasibility of the latter strategy.[23,24] Reactive pharmacogenetic testing refers to genotyping performed after a decision to prescribe a drug is made or to explain an unexpected drug response. These tests usually focus on the single gene specific for that drug.[24] Pre-emptive pharmacogenetic testing, on the contrary, occurs prior to any prescribing decisions.[24] In this scenario, patients are genotyped for a panel of multiple pharmacogenes simultaneously, so that broad pharmacogenetic results are readily available in electronic health records to guide future prescribing. Indeed, the clinical implementation

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indicate a potential impact on drug safety or pharmacokinetics, respectively. As of late 2022, 11 cardiovascular drugs are listed throughout the table, 3 of which are included in section 1 (clopidogrel, propafenone, and warfarin). The "Biomarkers" Table shows that 17 cardiovascular drugs (including rosuvastatin) currently have pharmacogenomics information provided in their drug label. However, not all drugs listed in the "Biomarkers" Table are also contained in the "Associations" Table, and vice versa. Additionally, some pharmacogenomic information in drug labels may be outdated. For example, the warfarin label only provides dosing recommendations based on *CYP2C9 *2* and *3* and does not acknowledge the additional alleles important in other ethnic groups (see warfarin section below). Moreover, even if a drug is listed in the "Biomarkers" Table, that does not necessarily mean that clinical action is recommended by the FDA. For example, rosuvastatin and *SLCO1B1* are listed in the "Biomarkers" Table, but the mention of pharmacogenetics is only informational, and no clinical action is recommended in the rosuvastatin label (i.e., "The impact of this polymorphism [*SLCO1B1*] on efficacy and/or safety of rosuvastatin has not been clearly established"). Due to rather vague therapeutic recommendations provided in cardiovascular drug labels, inconsistencies between the FDA tables, and lack of citations accompanying clinical study references, these FDA tables may be considered a secondary resource useful in instances where a CPIC guideline is unavailable for the gene-drug pair in question (e.g., we would recommend the FDA tables for propafenone, which is not covered by CPIC guidelines).

American Heart Association/ American College of Cardiology

The AHA and the ACC are the leading resources in the United States for clinical practice recommendations for cardiovascular diseases, but neither have a centralized repository for pharmacogenetic recommendations or guidelines. This is likely because neither provides much pharmacogenetic guidance in general. As of late 2022, keyword searches for "pharmacogenetics" and "pharmacogenomics" at the Library of Guidelines and Clinical Documents on the ACC website (https://www.acc.org/guidelines) and the Guidelines and Statements Search on the AHA website (https://professional.heart.org/en/guidelines-and-statements/guidelines-and-statements-search) yield very limited results. The ACC *Recommendations for Beta Blockers and Amiodarone in Patients Undergoing CABG* from 2021 states, "Beta-blocker pharmacogenetic variation may have a role" in mortality rates after coronary artery bypass graft surgery. When filtering the "pharmacogenetics" or "pharmacogenomics" AHA search to only "Clinical Practice Guideline" document types, three articles are returned but only the *Guidelines for the primary prevention of stroke* (2014) has guidance, stating "Pharmacogenetic dosing of vitamin K antagonists may be considered when therapy is initiated." Although there is a paucity of guidelines with pharmacogenetic recommendations from these organizations, the AHA released a scientific statement in 2016 stating, There are specific drugs for which pharmacogenetic variant information can be reasonably used today, and refers readers to the FDA pharmacogenetic tables, PharmGKB, and CPIC.[58] However, unlike guidelines, scientific statements from the AHA do not undergo the same rigorous and transparent process for evidence evaluation, and clinical recommendations are technically not allowed in AHA scientific statements. Due to the lack of a centralized resource for pharmacogenetic-specific recommendations, outdated content, and difficulty searching/retrieving the documents that are available, we recommend

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Statins

Statins are mostly well-tolerated, although some patients report statin associated musculoskeletal symptoms (SAMS). The SAMS refers to a range of disorders from the rare rhabdomyolysis (1 in 10,000) to muscle pain with evidence of muscle damage or myopathy (1 in 2,000) to muscle symptoms without evidence of damage or myalgia (up to 1 in 10).[55] The *SLCO1B1* gene encoding the hepatic transporter OATP1B1 is the most validated gene underlying the risk for statin myopathy. First identified through a genomewide association study of myopathy in patients treated with simvastatin 80 mg,[64] the reduced function haplotype (*5) has since been extended to other SAMS phenotypes[65] and statins.[66] Beyond symptoms, the *5 variant has also been associated with premature statin discontinuation for simvastatin[67] and atorvastatin.[68] Importantly, SAMS risk (in *5 allele carriers) appears to be a class effect pharmacokinetically, although the strength of association is statin type- and dose-dependent.[65] Additionally, there is recent evidence of a differential effect of *SLCO1B1* genotype by sex requiring further investigation.[69] The biggest updates to the most recent CPIC guidelines for statins (published in 2022), compared with the 2014 update, is the expansion of statin types and genotypes beyond only simvastatin and *SLCO1B1*, respectively (Table 3).[55,70] Designed to mirror the 2018 multi-society guidelines for managing blood lipids,[71] the 2022 CPIC guidelines for statins begin with an assessment of the patient's cardiovascular risk and desired statin lipid lowering intensity (low, moderate, or high). Based on this assessment and the patient's SLCO1B1 phenotype (reduced or poor function), the guidelines provide statin types/doses and their associated SAMS risk. For example, a provider assessing a patient who is an *SLCO1B1* reduced function carrier with established atherosclerotic cardiovascular disease requiring high-intensity statin therapy should avoid atorvastatin 80 mg because of its high risk of SAMS. Instead, rosuvastatin 20 mg would be preferred to achieve a similar LDL-C lowering effect. The statin CPIC guidelines are not intended to replace the 2018 multi-society guidelines.[55] Beyond *SLCO1B1*, there is evidence that genetic variation in *CYP2C9* is associated with increased exposure to fluvastatin.[72] Moreover, variation in *ABCG2*, which encodes an efflux transporter (BCRP) that modulates the absorption and disposition of rosuvastatin, is associated with rosuvastatin exposure.[73] Accordingly, the CPIC guidelines include recommendations for dosing rosuvastatin based on *ABCG2* phenotype in combination with *SLCO1B1*, and for dosing fluvastatin based on *CYP2C9* phenotype. With this recent expansion of CPIC guidelines to commonly prescribed high-intensity statins (i.e., atorvastatin and rosuvastatin), we anticipate increased use of pharmacogenetic-guided statin selection and dosing in clinical care. The CPIC guideline recommendations are based on level A strength of evidence.[55] Additionally, the FDA has recently determined that clinical confirmatory testing is no longer required for the 23andMe DTC pharmacogenetic test of *SLCO1B1*-simvastatin.[74]

Warfarin

The 2017 CPIC guidelines for genotype-guided warfarin therapy focuses on dose requirements using validated published pharmacogenetic algorithms based on *CYP2C9*, *VKORC1*, *CYP4F2*, and rs12777823 genotype (level A evidence; Table 3). Variants in the *CYP2C9* and *VKORC1* genes contribute to decreased warfarin metabolism and altered sensitivity, and patients with these polymorphisms may need modified starting doses to